



GOAT: GO to Any Thing

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Third-person view



1. GOAT Problem

GOAT System Architecture

Results

Applications

Pick & Place

Social Navigation

Platform Agnostic

Unknown Environment
Explore

Perception
*Detect and Localize
Objects*

Lifelong Memory
*Remember Object
Locations*

Control
*Navigate to / Pick & Place
Objects*



Multimodal:

Reach Any Object Specified in Any Way

Image



Language

Find **the fruit basket** on the **kitchen counter**

Category

Bring me **a CUP**

Lifelong:

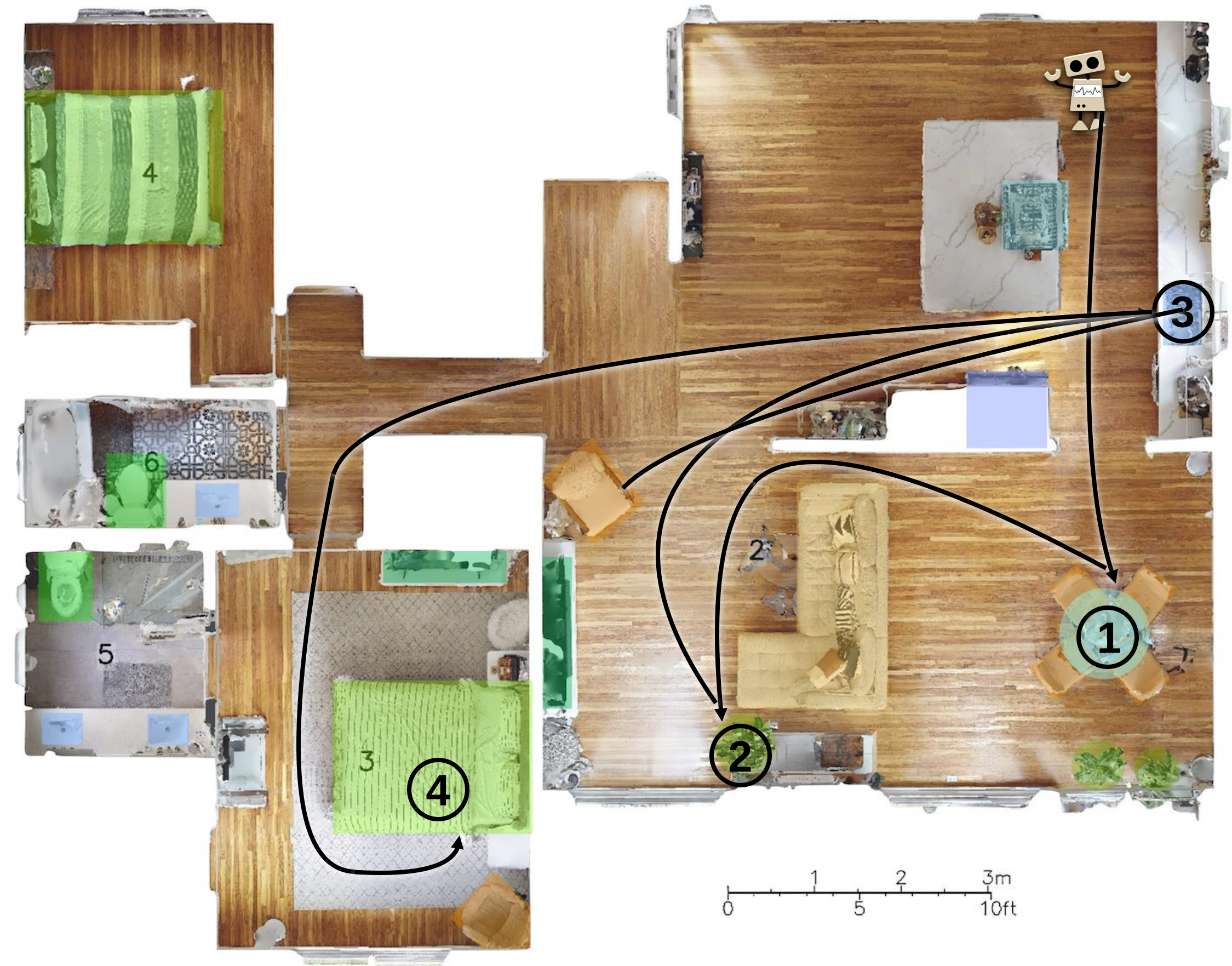
Remember Object Locations



② Go to the potted plant next to the couch

③ Go to a SINK

④ Go to the black and white striped bed



1. GOAT Problem

GOAT System Architecture

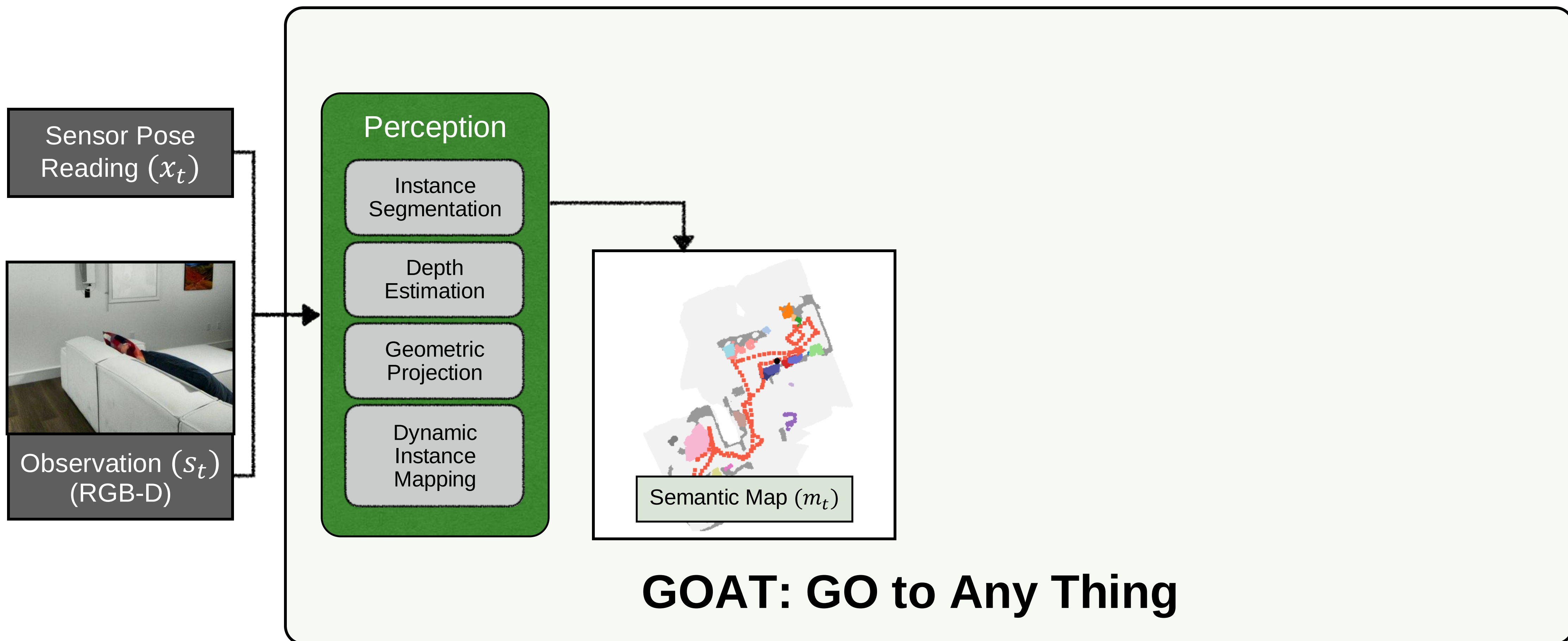
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Language Goal

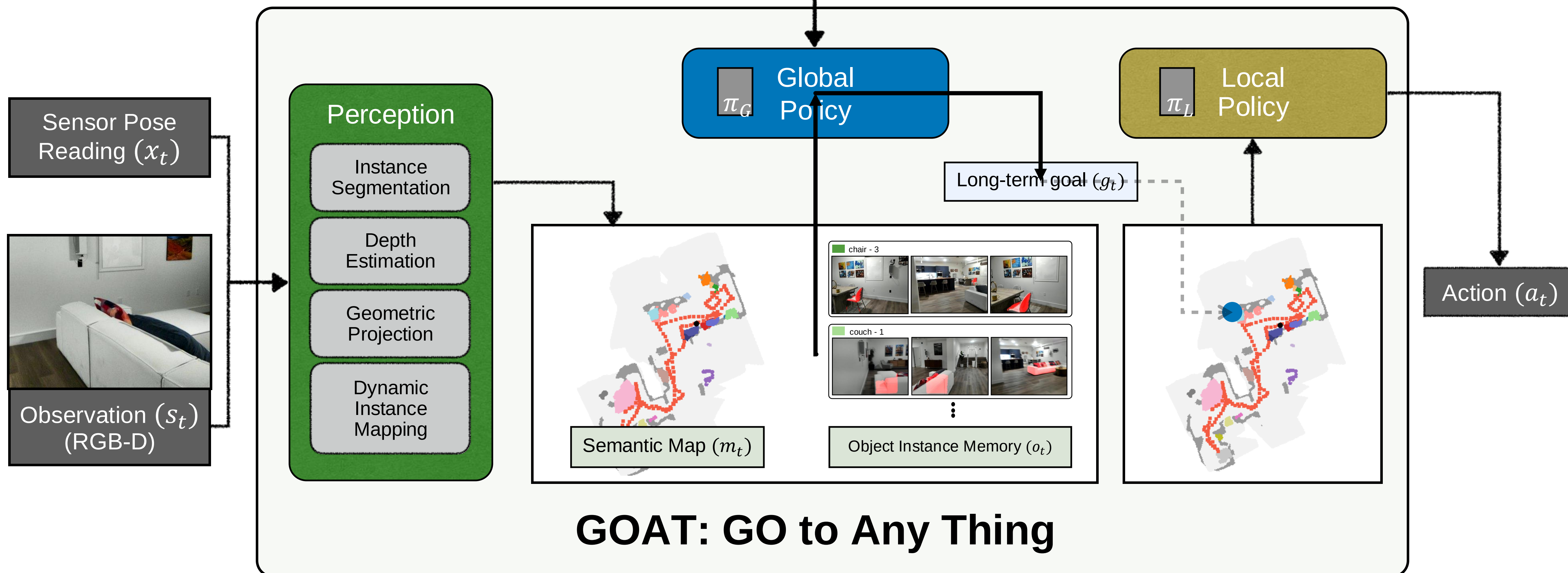
Brown office chair
Bed with white sheets
Cup on the kitchen counter

Image Goal

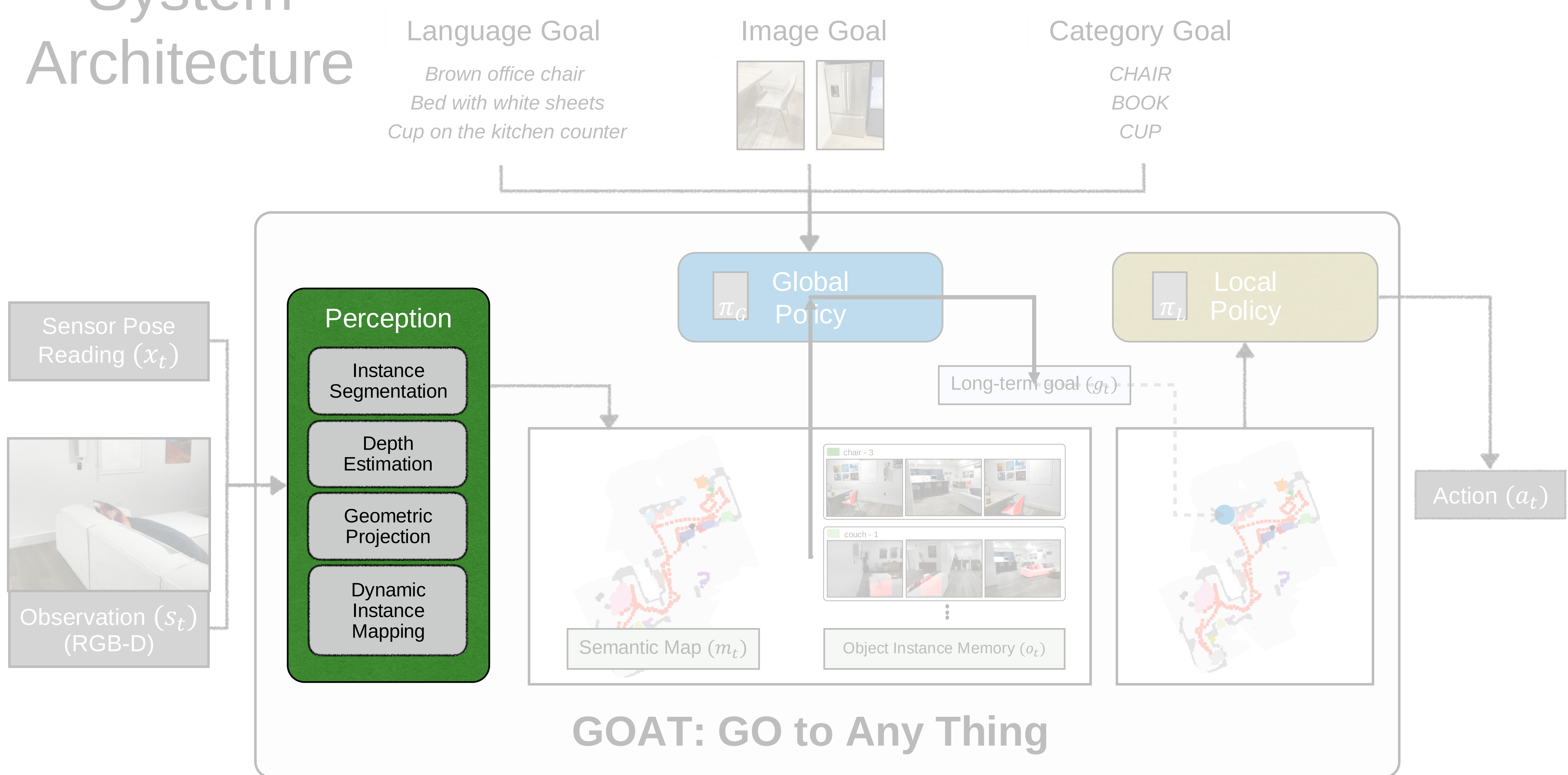


Category Goal

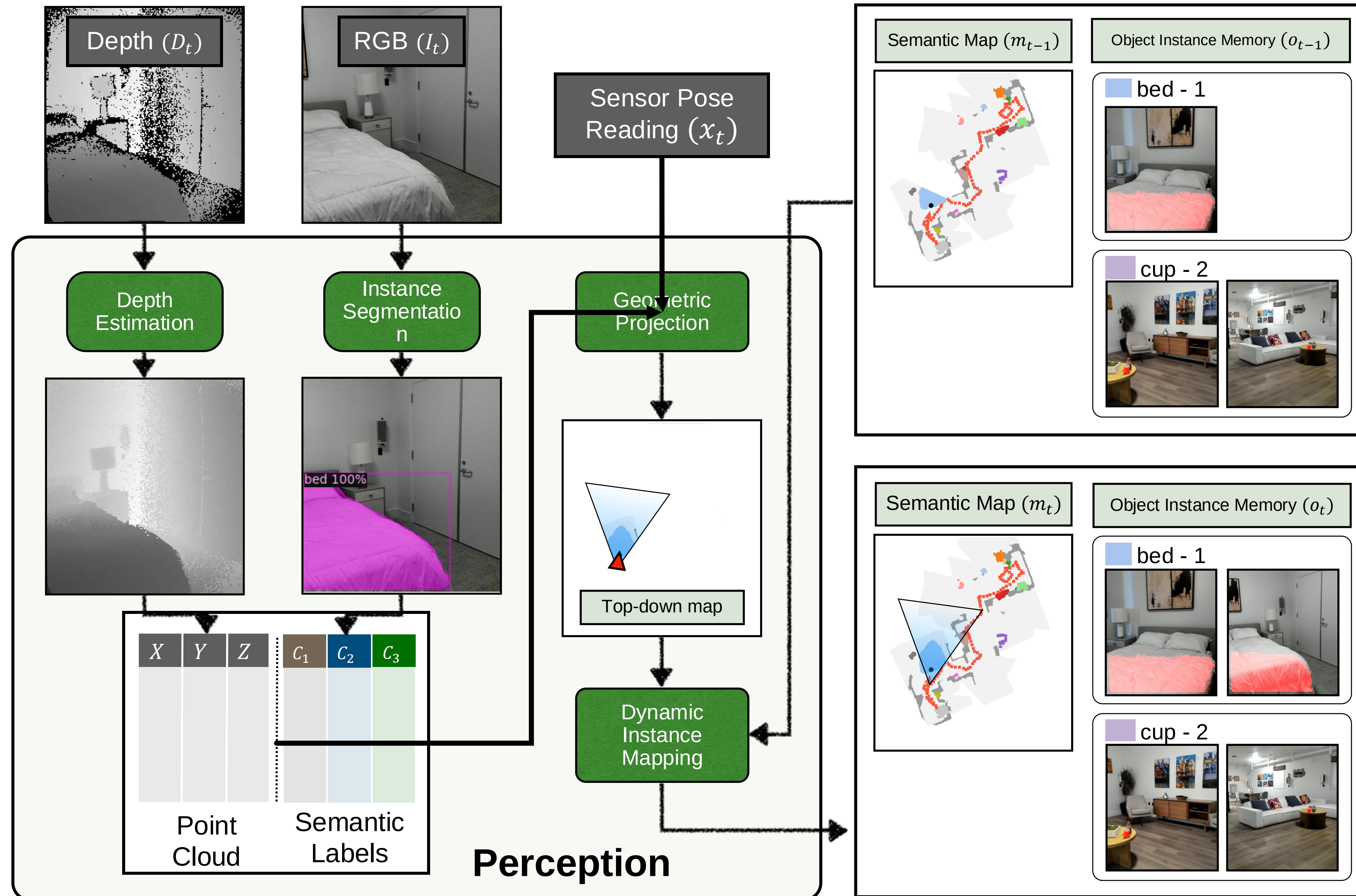
CHAIR
BOOK
CUP



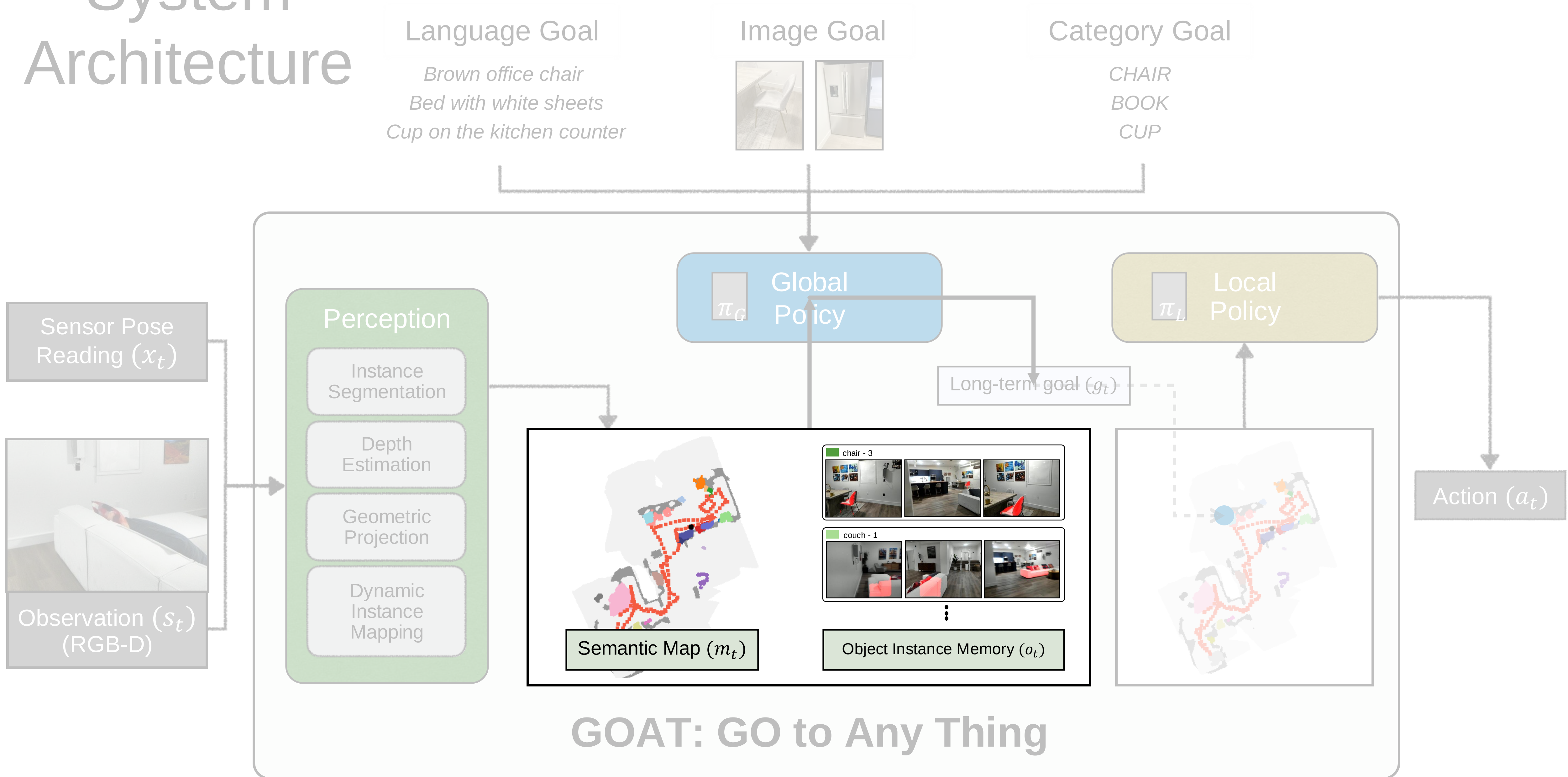
System Architecture



Perception System

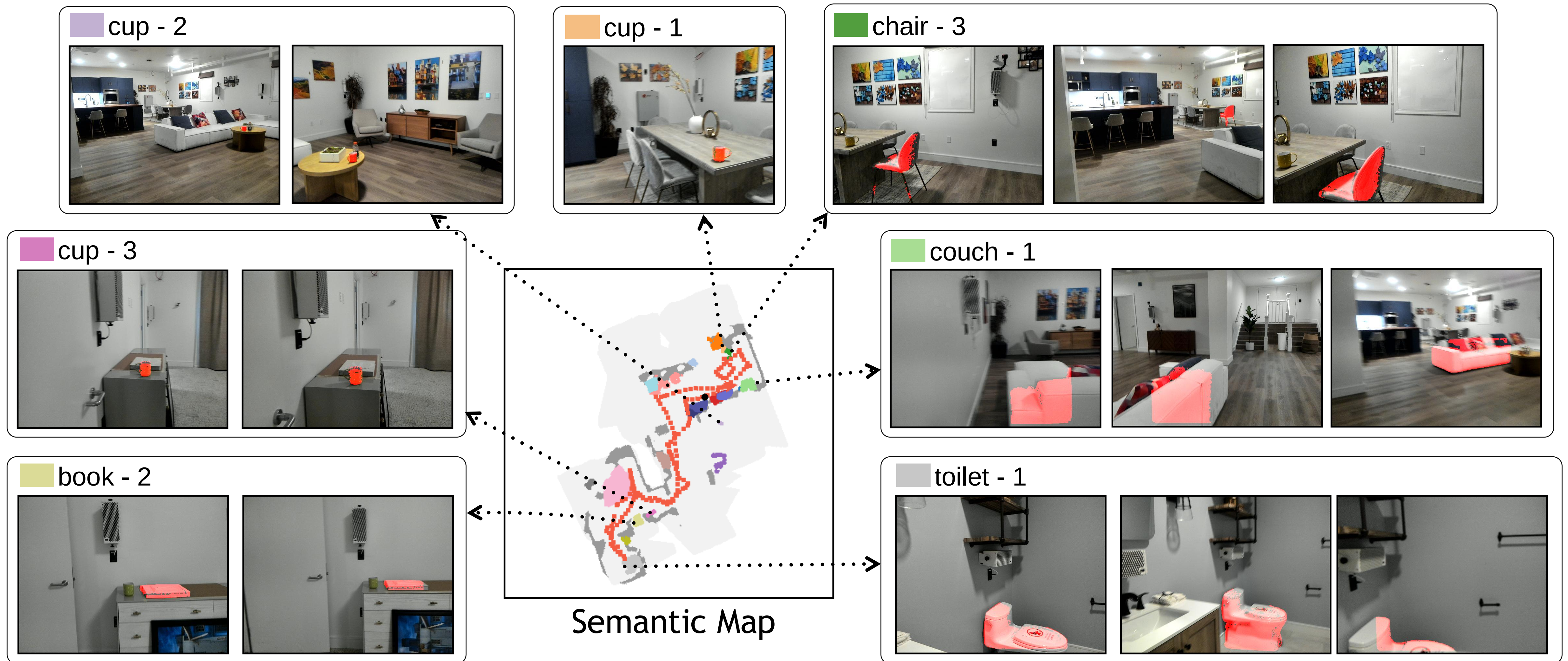


System Architecture



GOAT Memory Representation

Semantic map with associated Object Instance memory



Object Instance Memory

chair - 3



couch - 1



toilet - 1

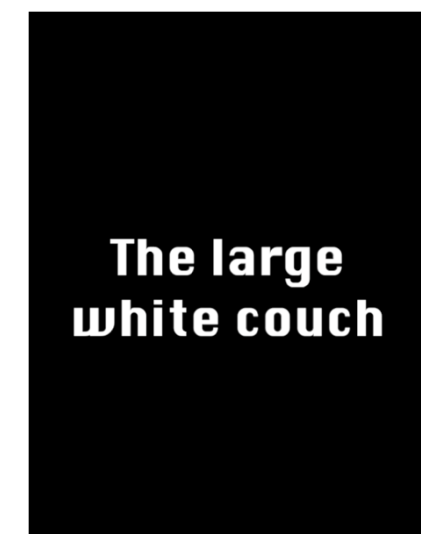


book - 2



⋮

Goals



Goal Localization

Goal Localization

Goal Localization

Localized

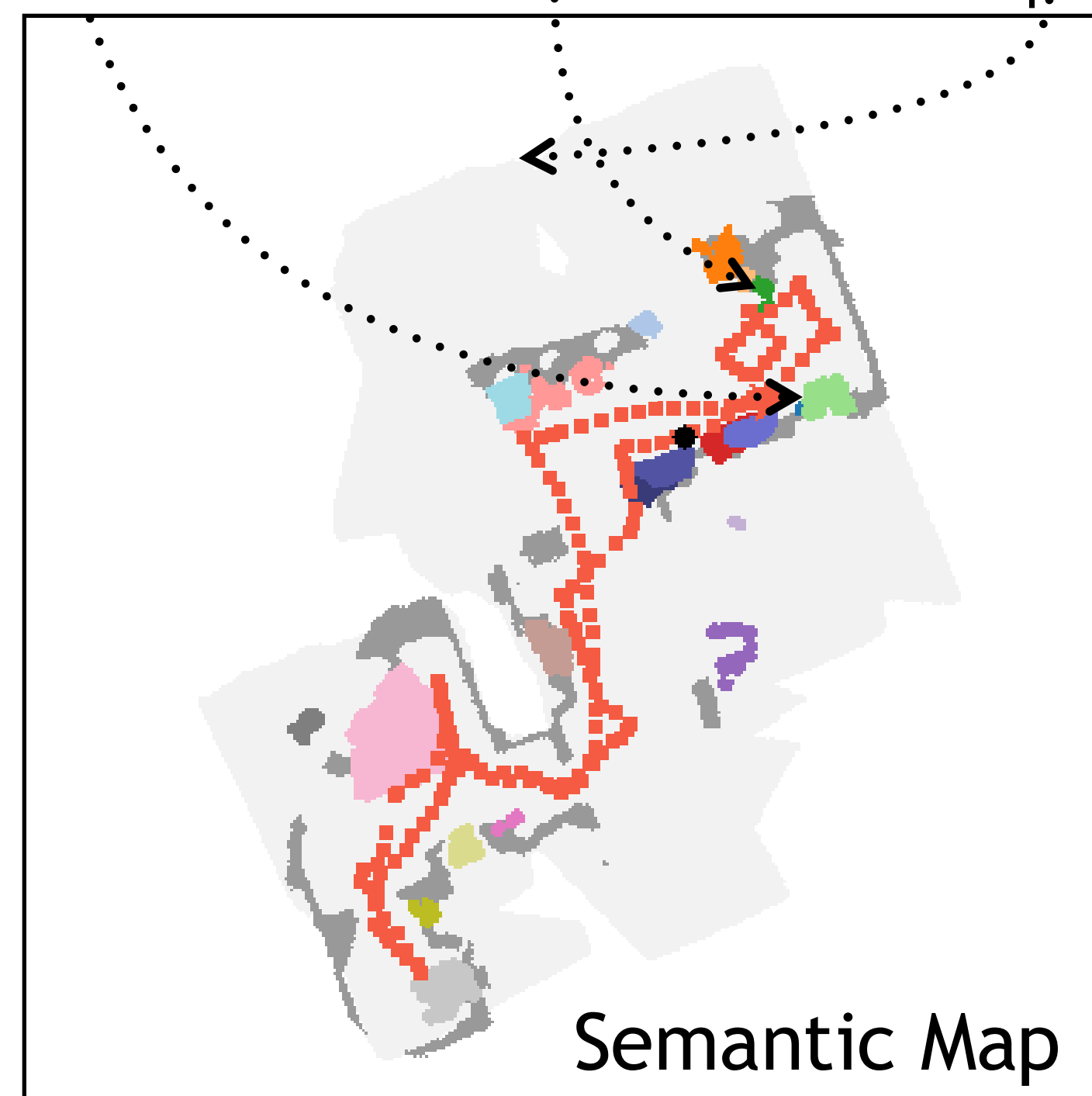
Localized

Not localized

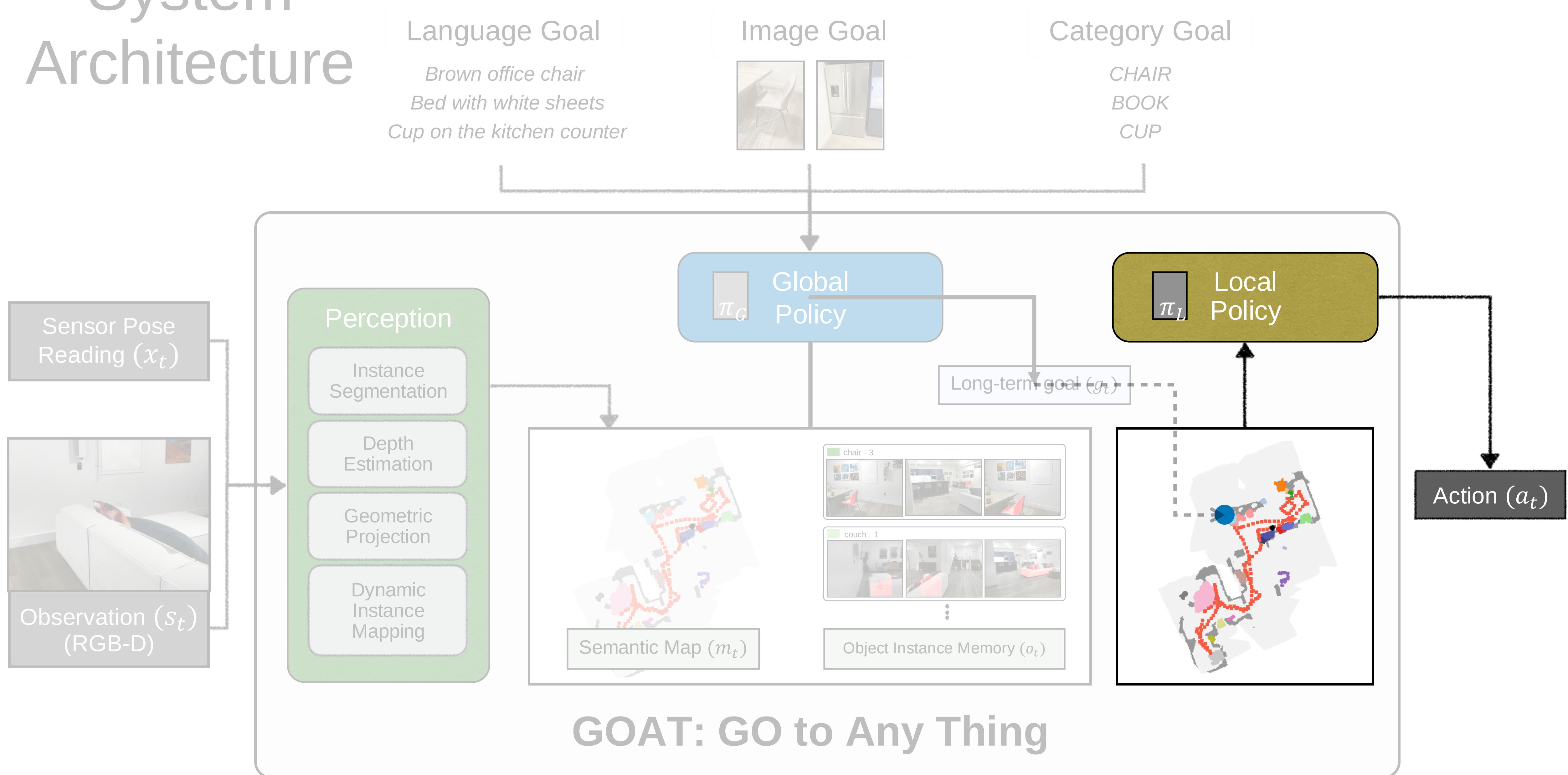
couch - 1

chair - 3

Explore



System Architecture



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“In the Wild”
Empirical Evaluation
9 Unseen Homes
4 Methods
10 Trajectories per Home
5-10 Goals per Trajectory
~90h of Experiments





The bed with the white blanket pulled back halfway and grey sheets.



The bunk bed with stars on the blanket.



The bed with blue and white sheets.



The with the blue and yellow painting above it and plant on the right.



The bed with the blue blanket and blue pillows.



The bed with grey sheets and light green trim.



The bed with the white blanket and red pillows



The large grey living room couch with many pillows.



The green couch.



The white rectangular couch with no pillows.



The large grey living room couch with many pillows.



The light brown couch with blue pillows.



The large grey couch in front of the yellow wall.



The couch covered in a white blanket.



The green cup on the kitchen counter



The red cup on the kitchen counter



The green mug on the plastic chair.



The grey cup on the nightstand.



The red cup on top of the glass table.



The red cup on the kitchen counter.



The light blue cup.



The plant in front of the window.



The potted plant next to the stairs.



The large potted plant in the bedroom.



The group of plants in front of the curtain.



The large potted plant next to the foosball table.



The large potted plant in front of the mirror.



The small potted plant on the hallway table



The grey dining table chair.



The green lawn chair.



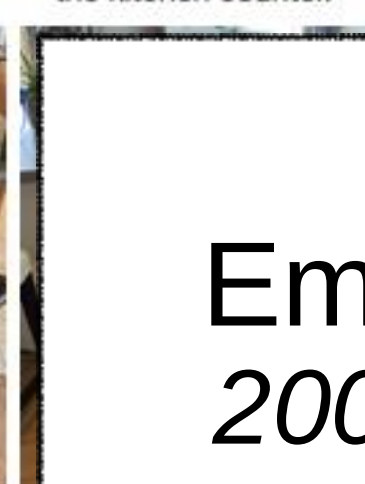
The black office chair.



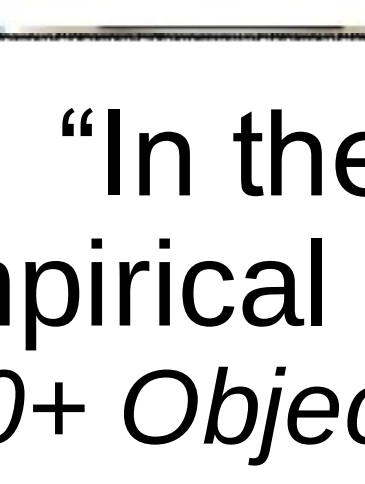
The small wooden chair at the desk.



The black leather chairs in the kitchen.



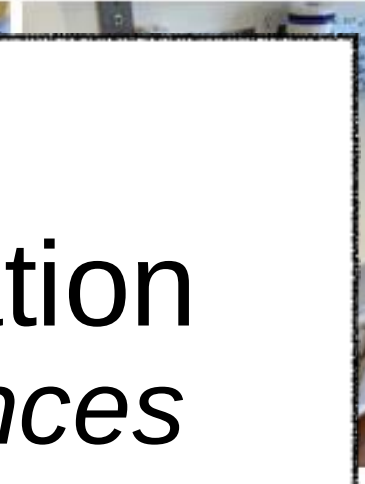
The bar-height chair at the kitchen island.



a wooden seat.



wood desk.



the kitchen counter

“In the Wild” Empirical Evaluation 200+ *Object Instances*



The light blue bowl on the living room coffee table.



The lavender bowl on the kitchen counter.



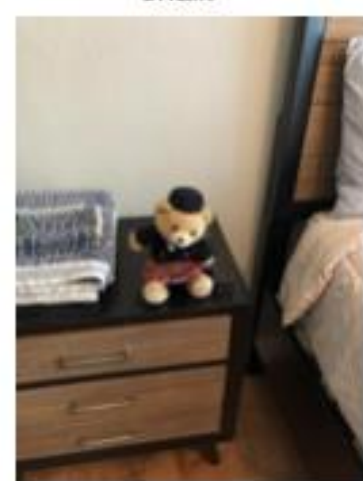
The light blue bowl on the couch.



The green cover book on the coffee table.



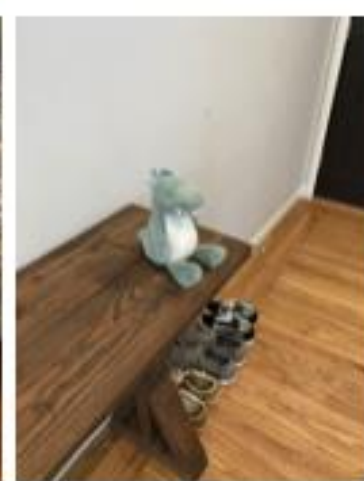
The book on the desk. It has a car on the cover.



The beige teddy bear.



The stuffed lion toy.



The green dinosaur stuffed toy.



The stuffed lion.



The beige teddy bear.



The stuffed lion toy.



The green dinosaur stuffed toy.



The refrigerator.



The refrigerator.



The refrigerator.



The refrigerator.



The toilet next to the shower curtain with blue fish.



The toilet.



The toilet next to the glass shower door.



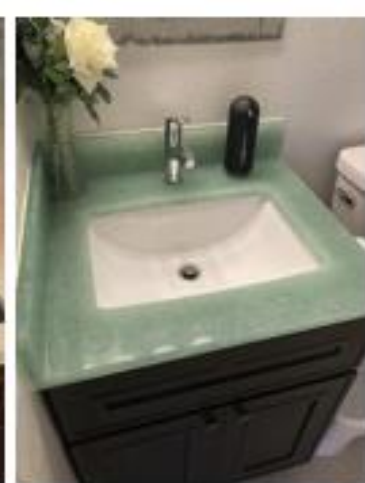
The kitchen sink.



The bathroom sink with marble top.



The kitchen sink.



The bathroom sink with green counter.



The bathroom sink.



The kitchen sink.



The bathroom sink with black and white counter.



The television.



The television mounted on a yellow wall.



The television mounted on a white wall.



The oven.



The oven.



The oven.



The oven.

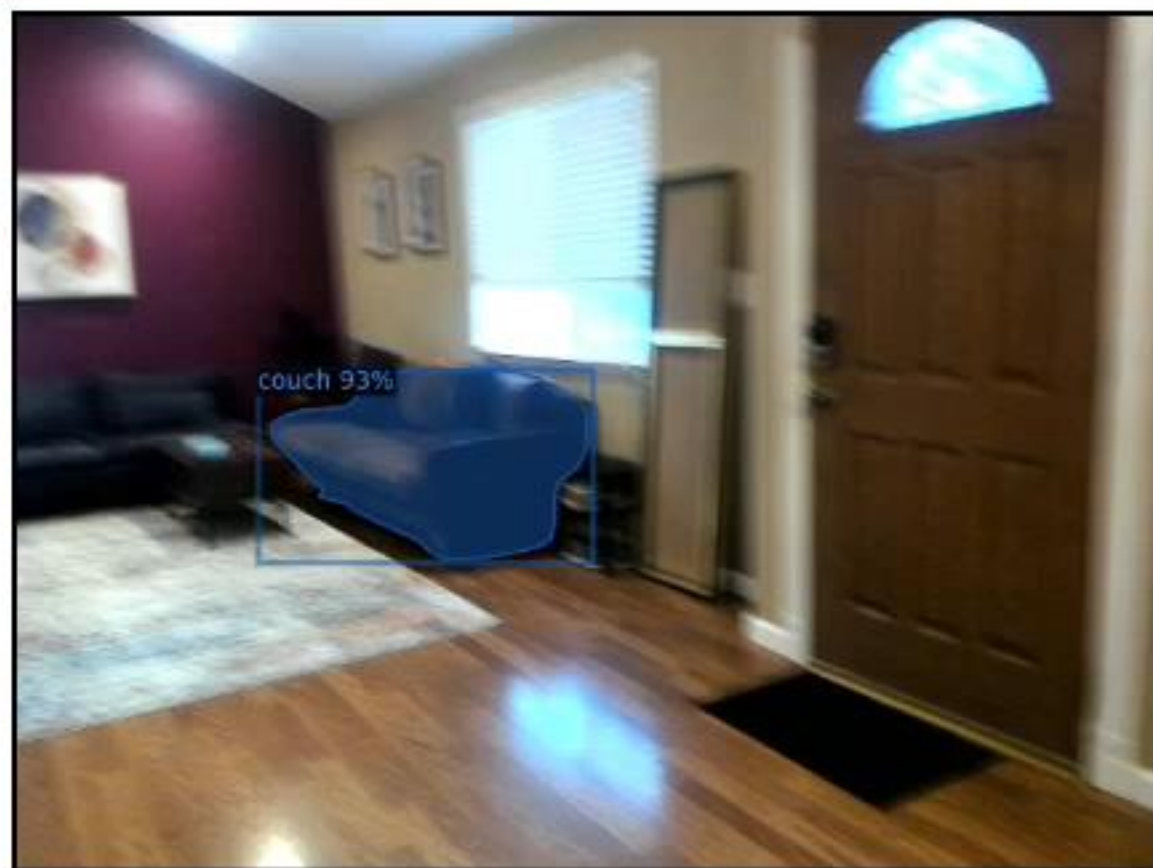
Observation



Third-person view



Observation



Third-person view



Semantic Map



couch - 1	chair - 1	chair - 2	couch - 2	chair - 3	chair - 4	chair - 5
refrigerator - 1	chair - 6	dining table - 1	potted plant - 1	potted plant - 2	sink - 1	oven - 1
bottle - 1	bottle - 2	chair - 7	tv - 1	chair - 8	teddy bear - 1	chair - 9
cup - 1						

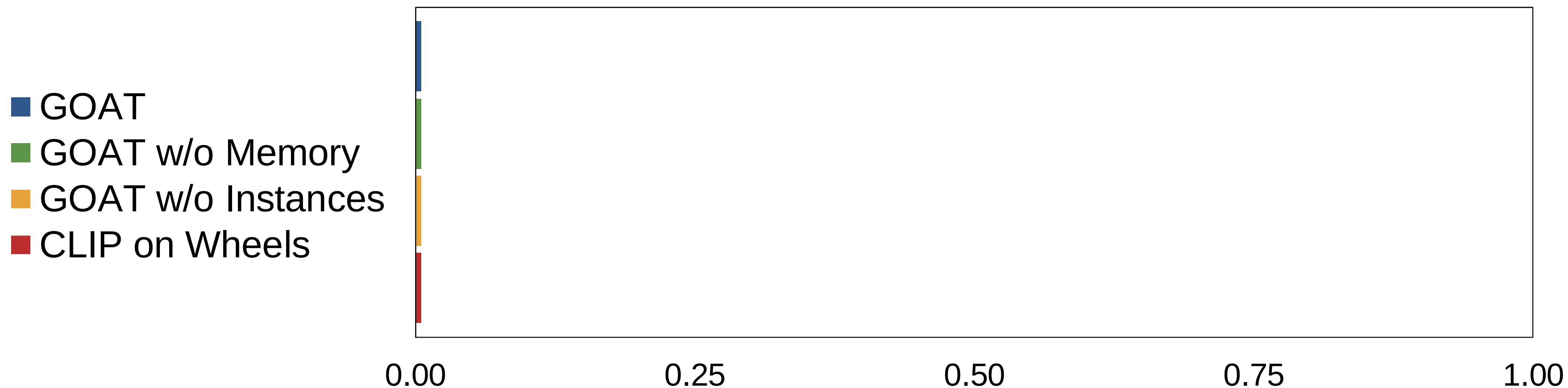
Third-person view



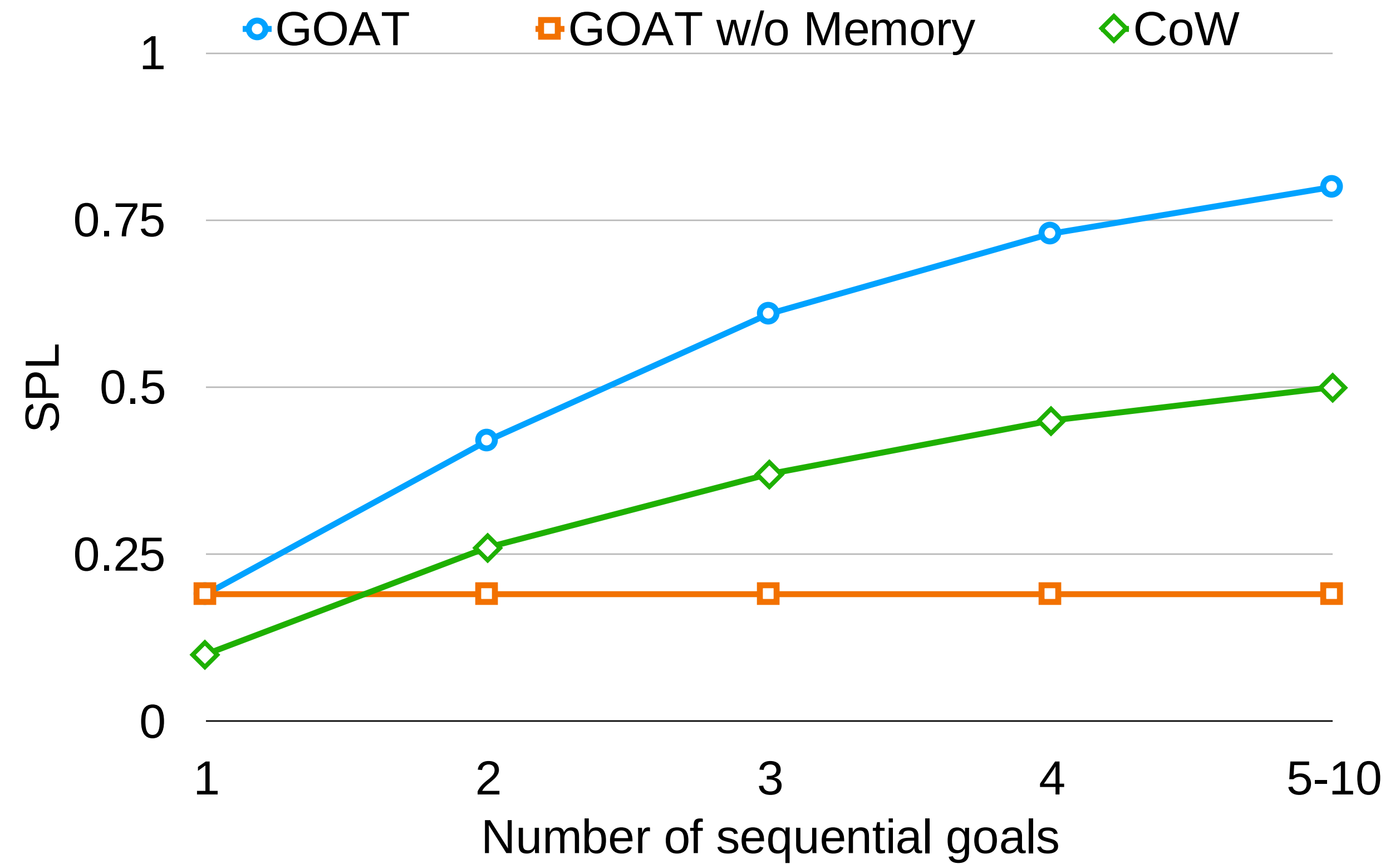
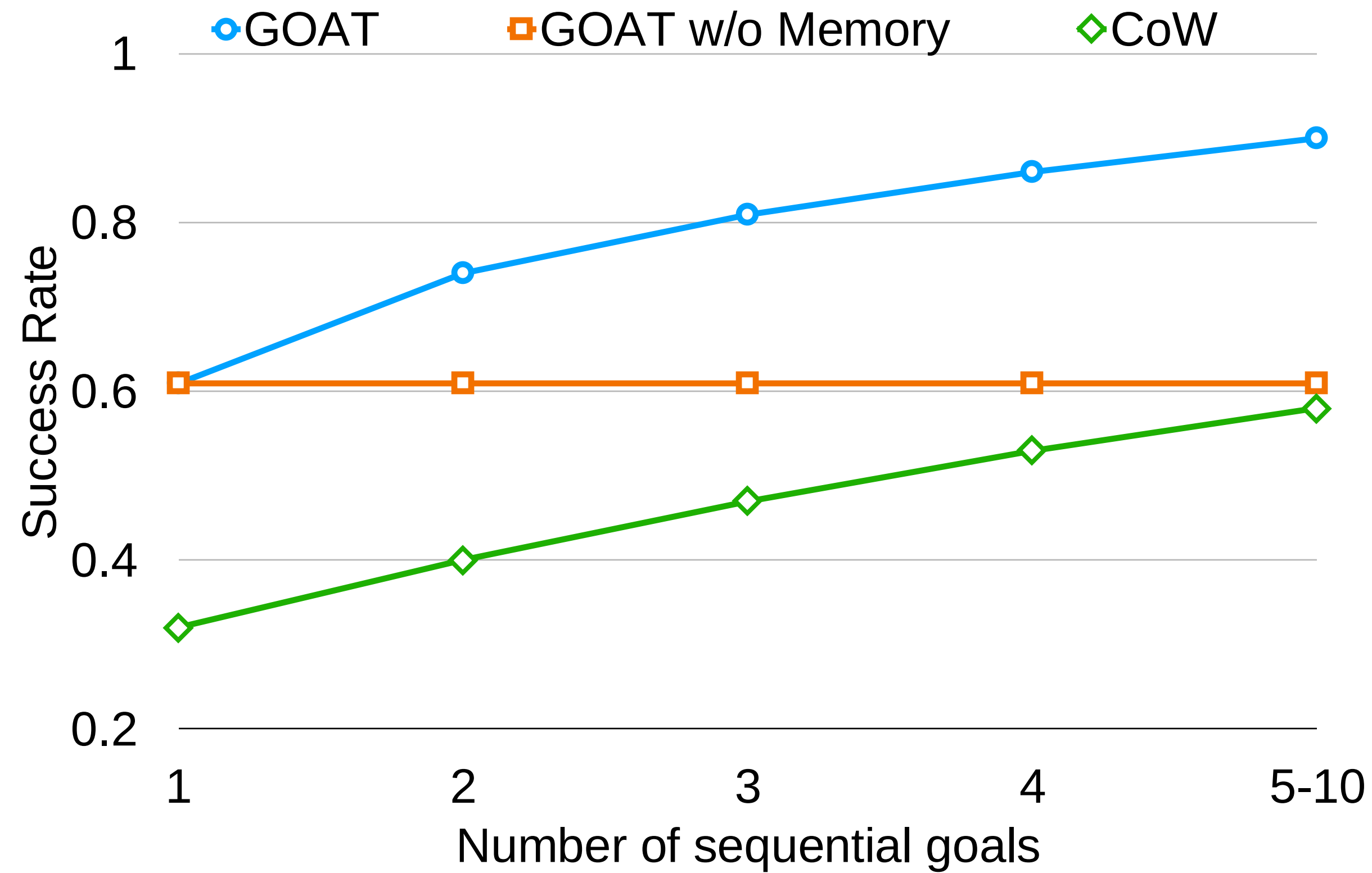
Results

Success Rate

SPL



Performance Across Episode



Goal:

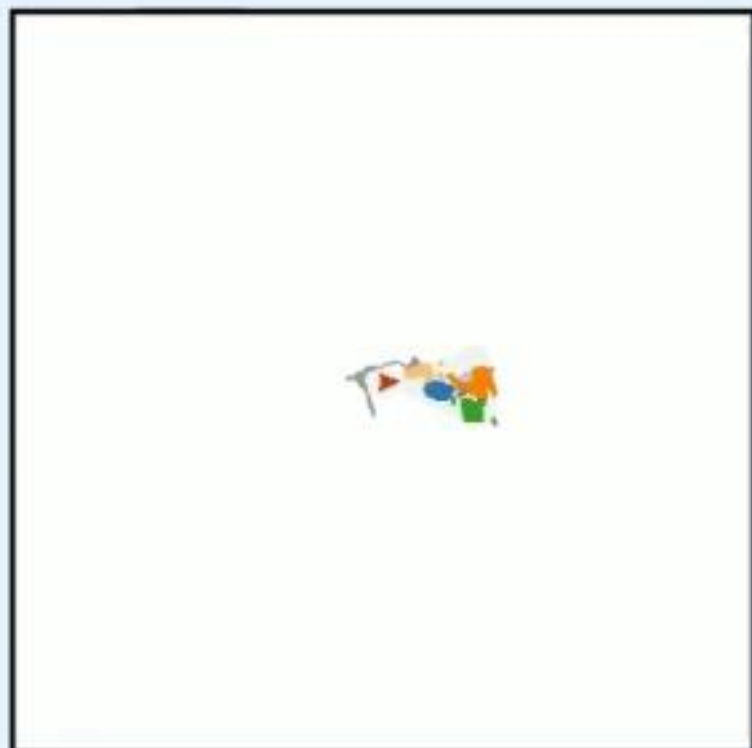


Baselines

Ours



Observation



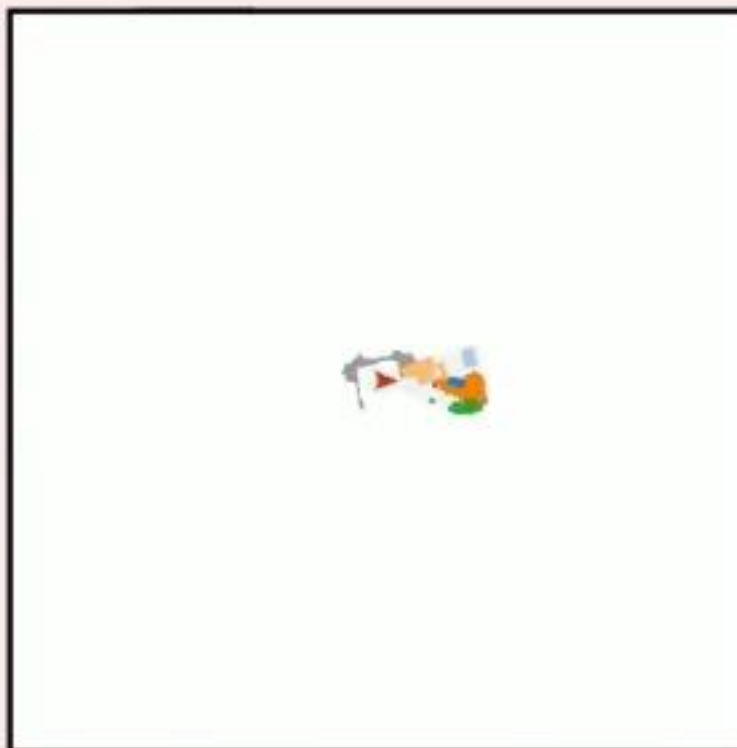
Instance Map

Success: 6/6 SPL: 0.78

No Memory



Observation



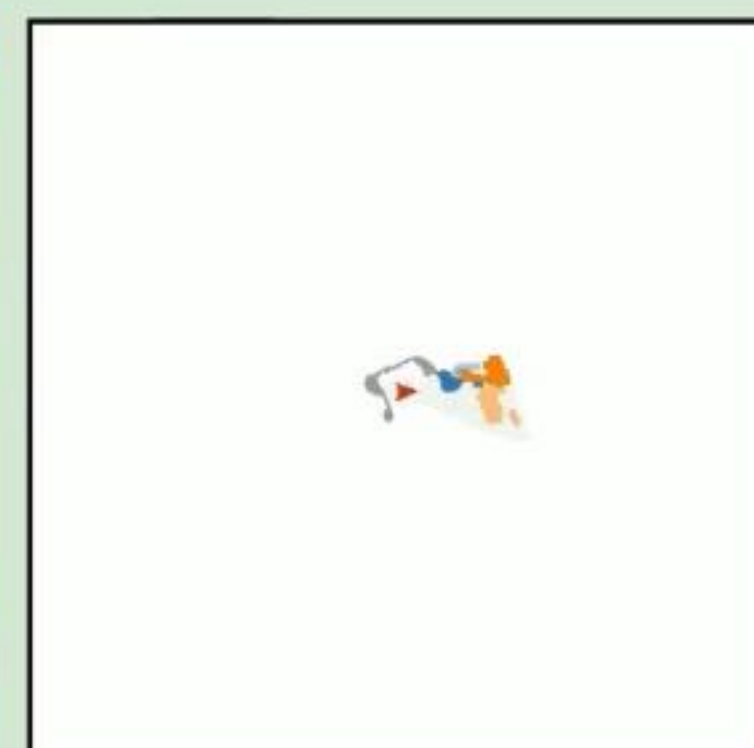
Instance Map

Success: 4/6 SPL: 0.40

CoW



Observation



Instance Map

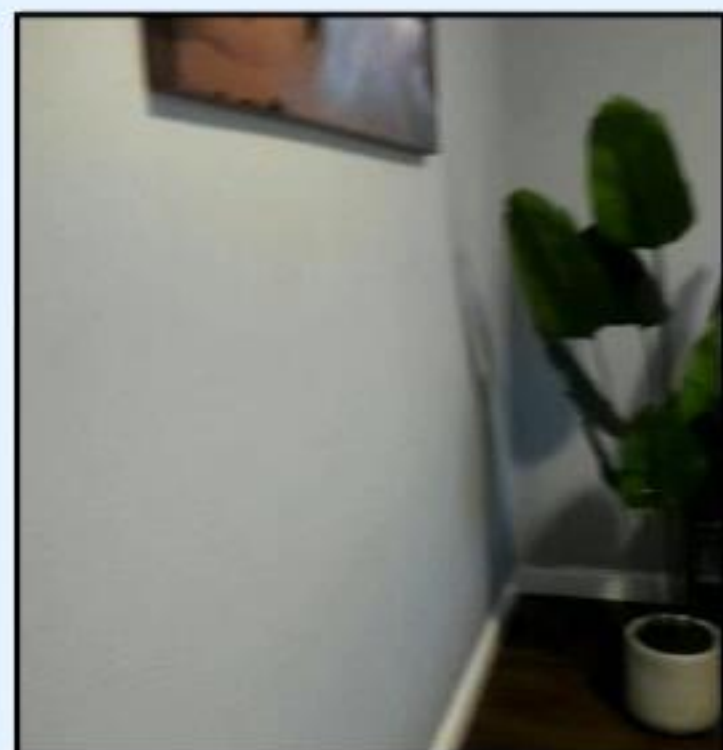
Success: 1/6 SPL: 0.16

Goal:



Baselines

Ours



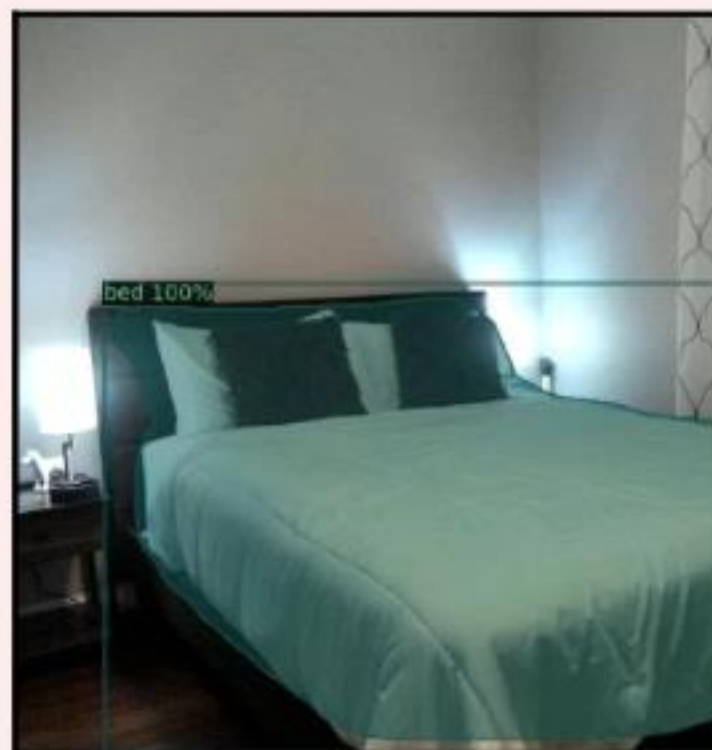
Observation



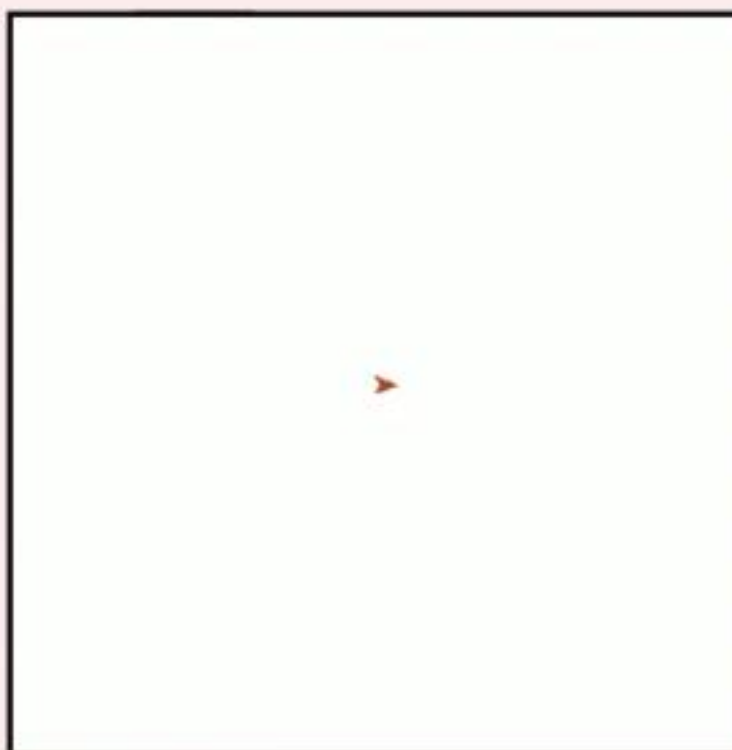
Instance Map

Success: 6/6 SPL: 0.78

No Memory



Observation



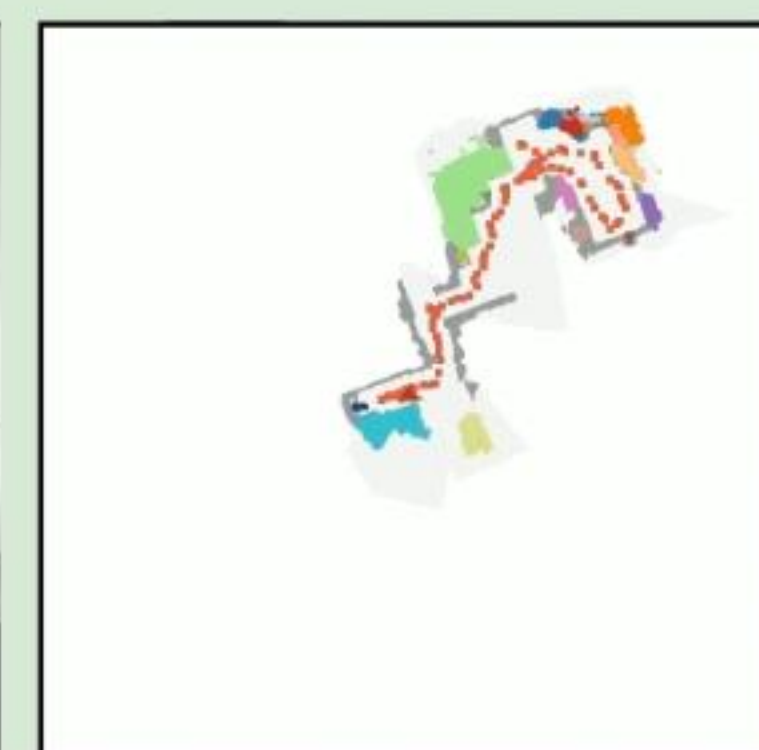
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CoW



Observation



Instance Map

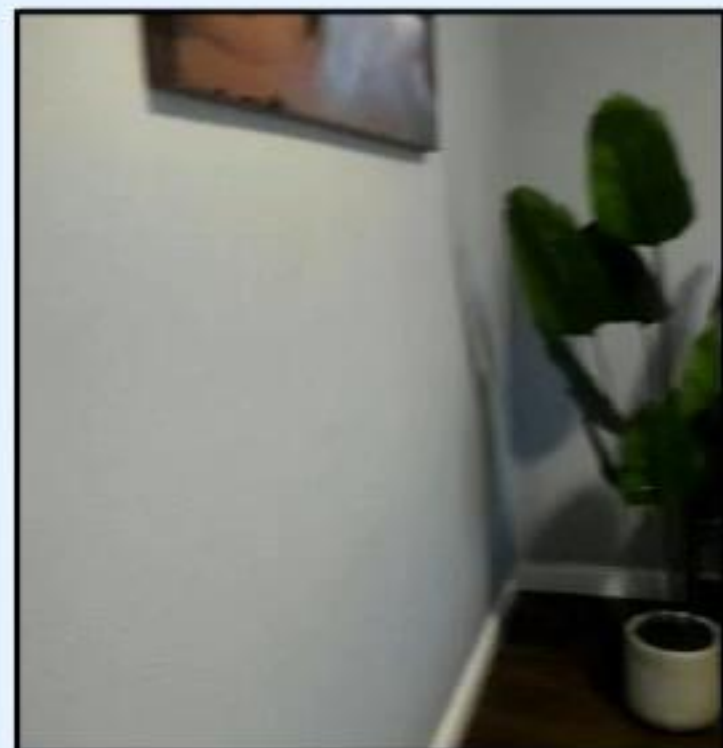
Success: 1/6 SPL: 0.16

Goal:



Baselines

Ours



Observation



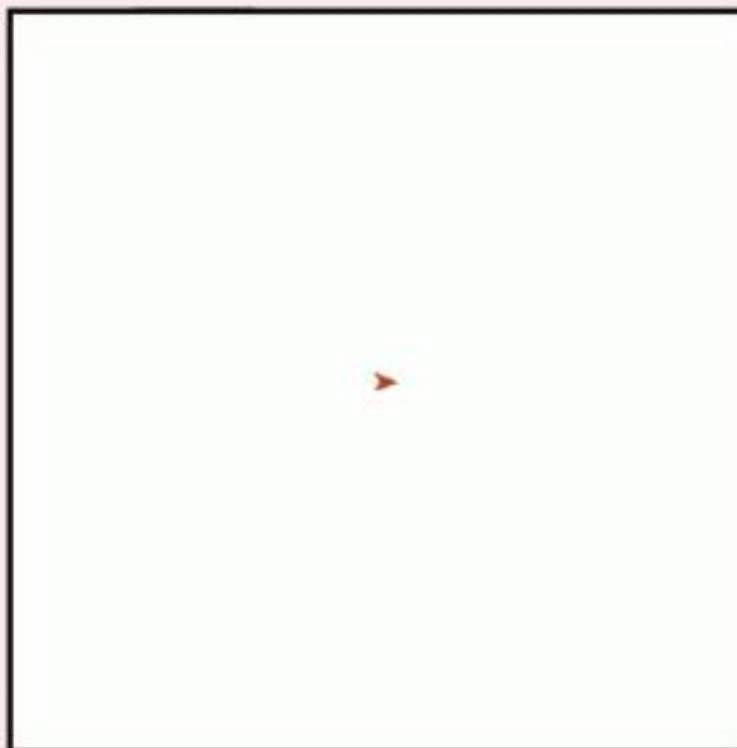
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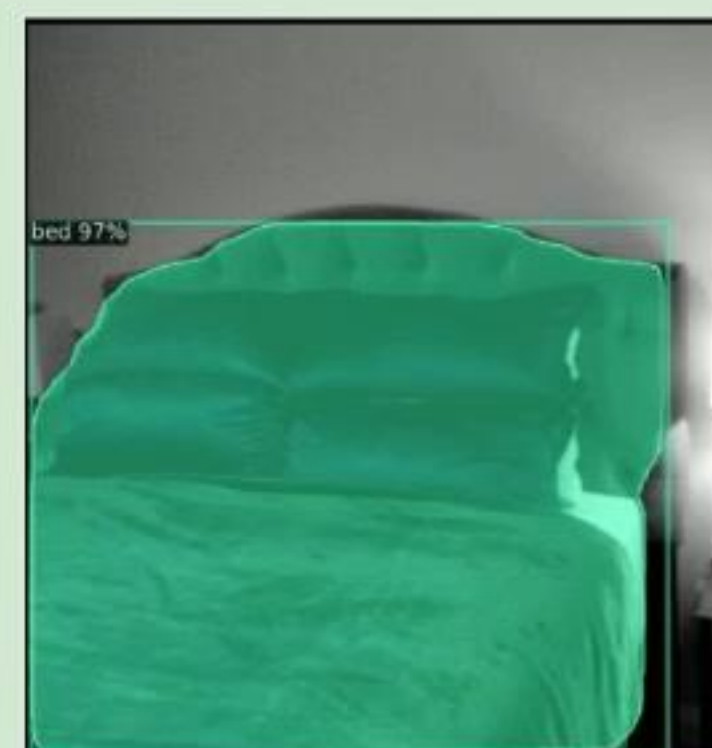
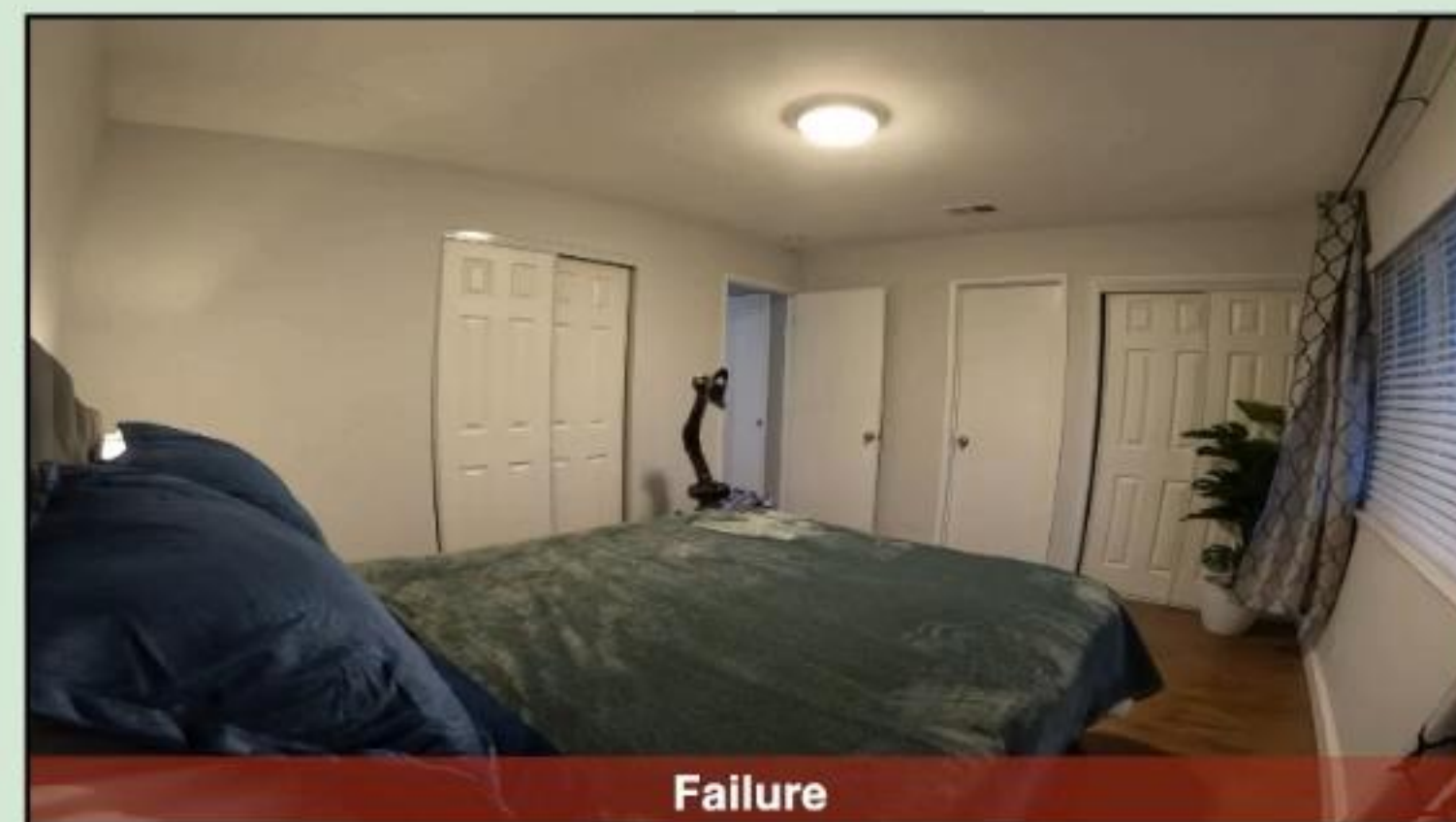
Observation



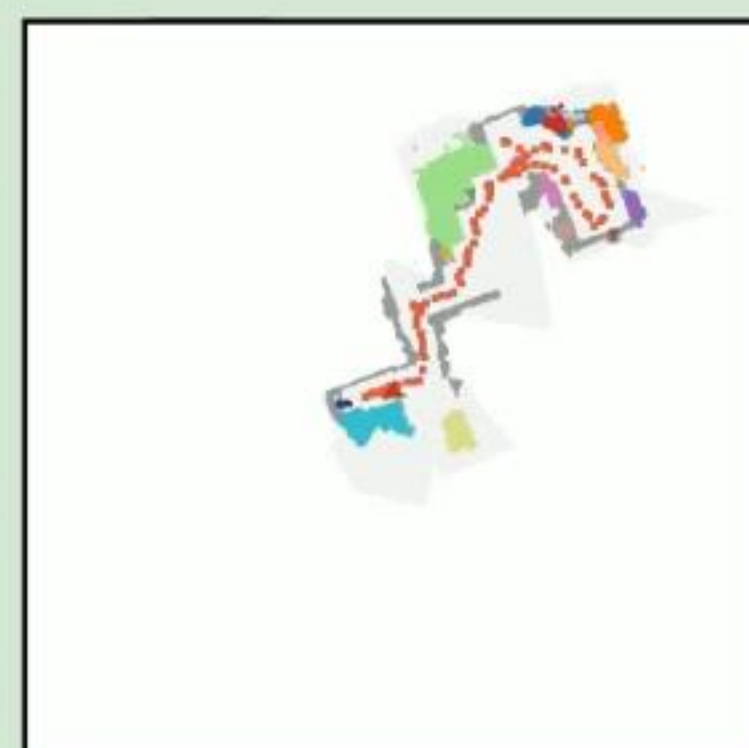
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CoW



Observation



Instance Map

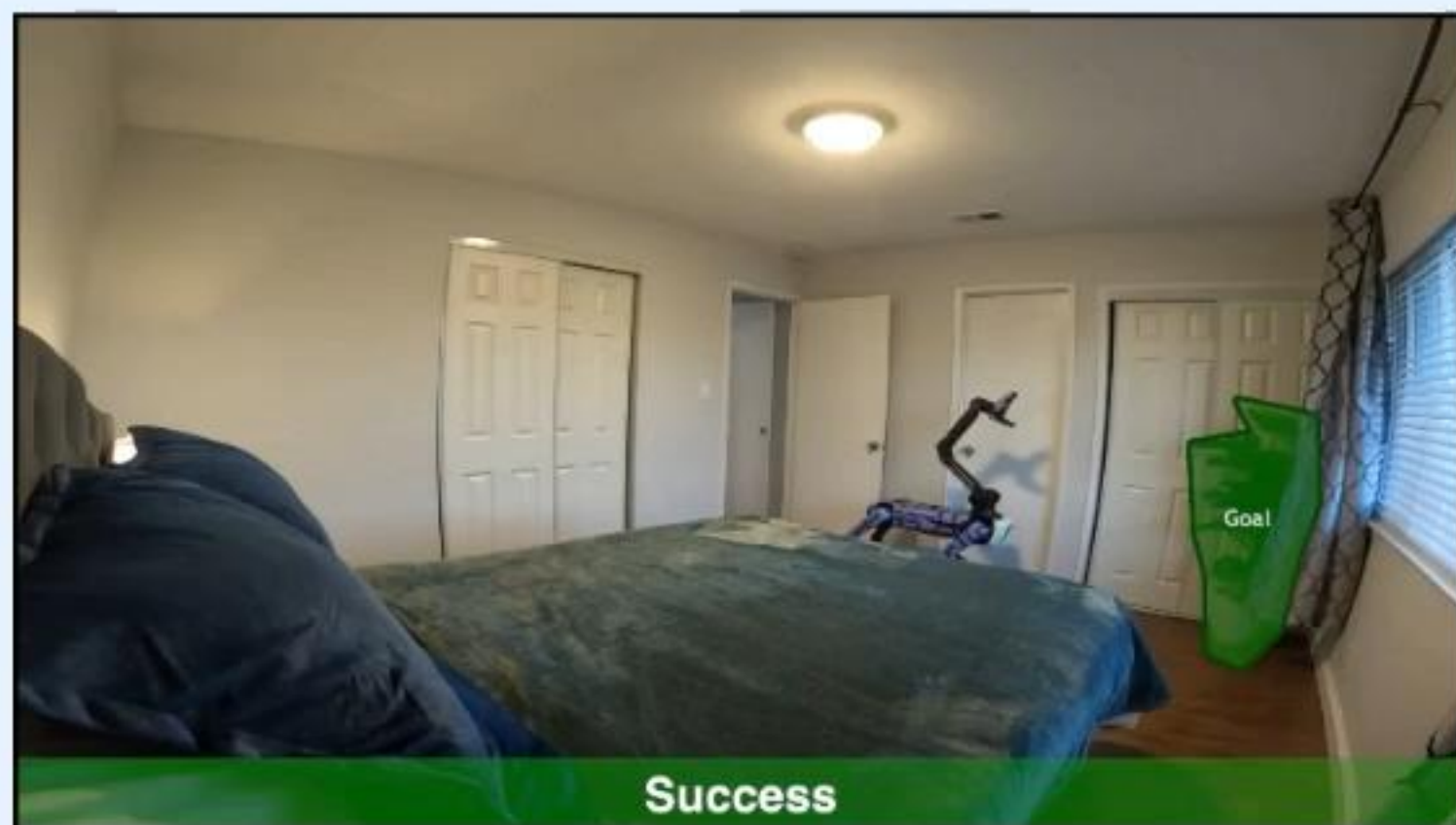
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Goal:

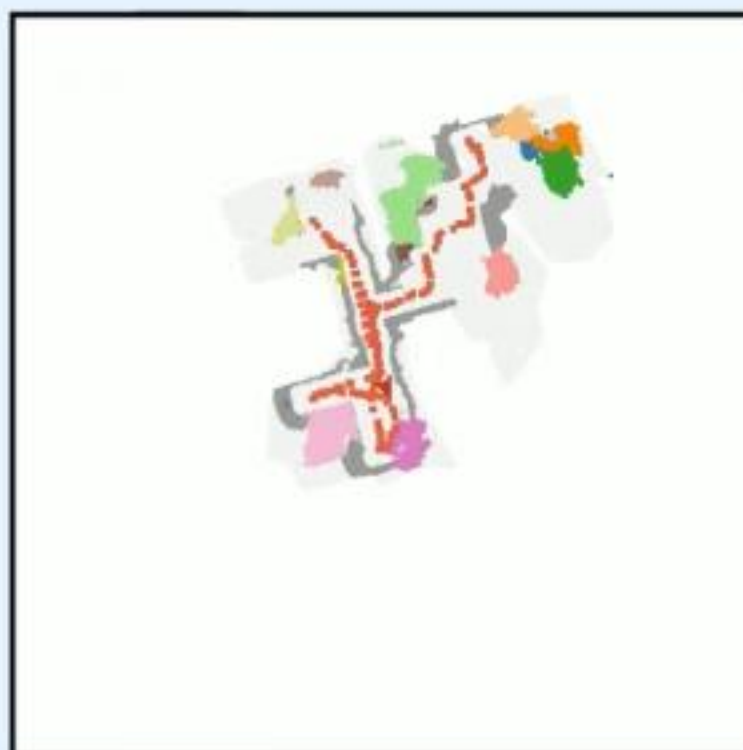


Baselines

Ours



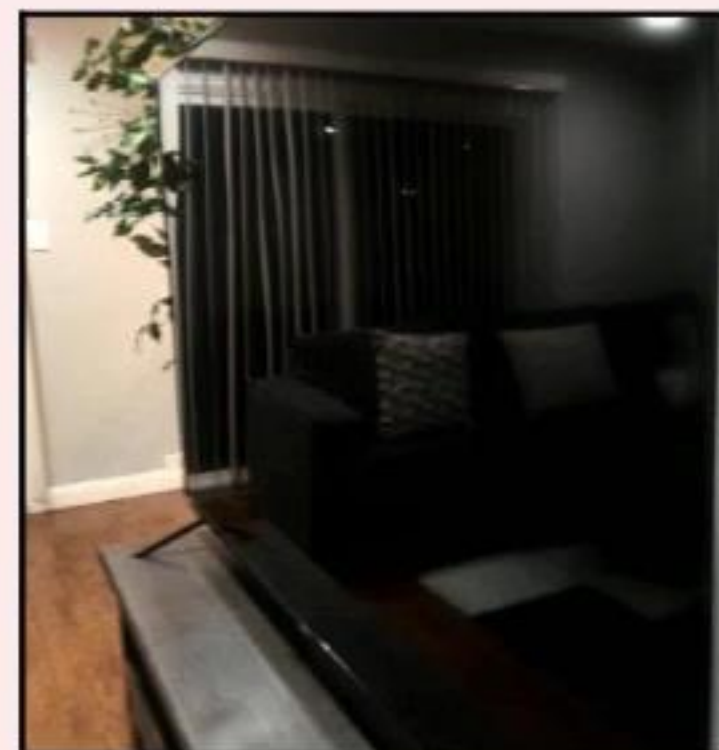
Observation



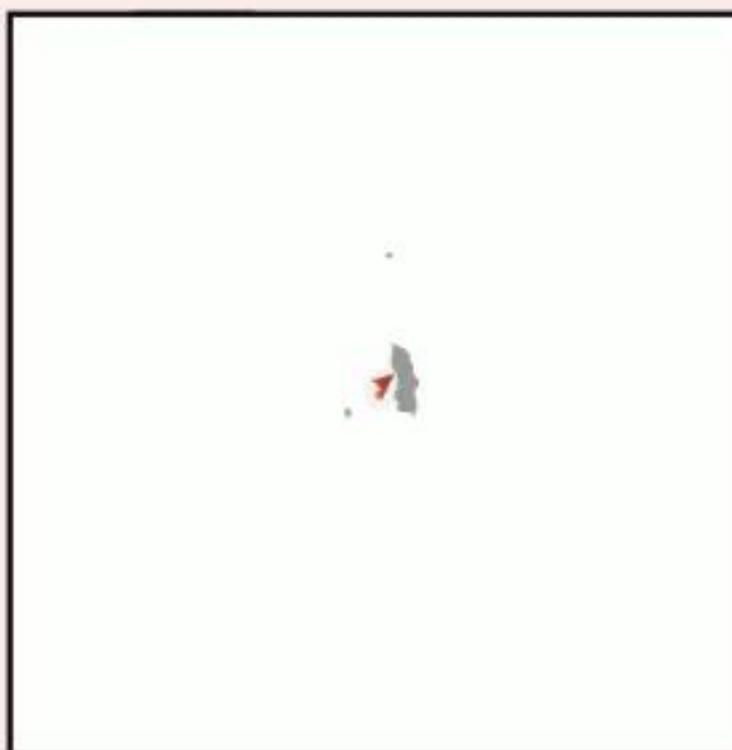
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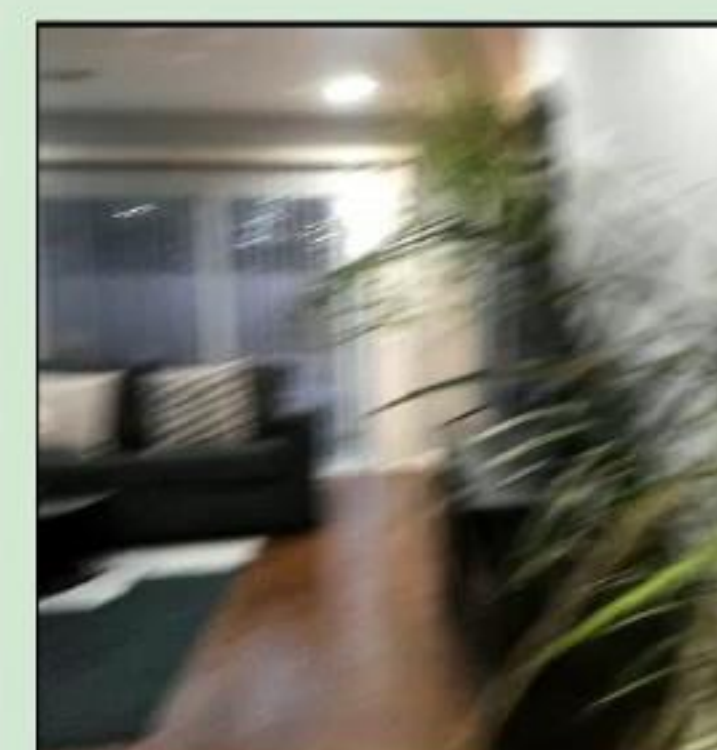
Observation



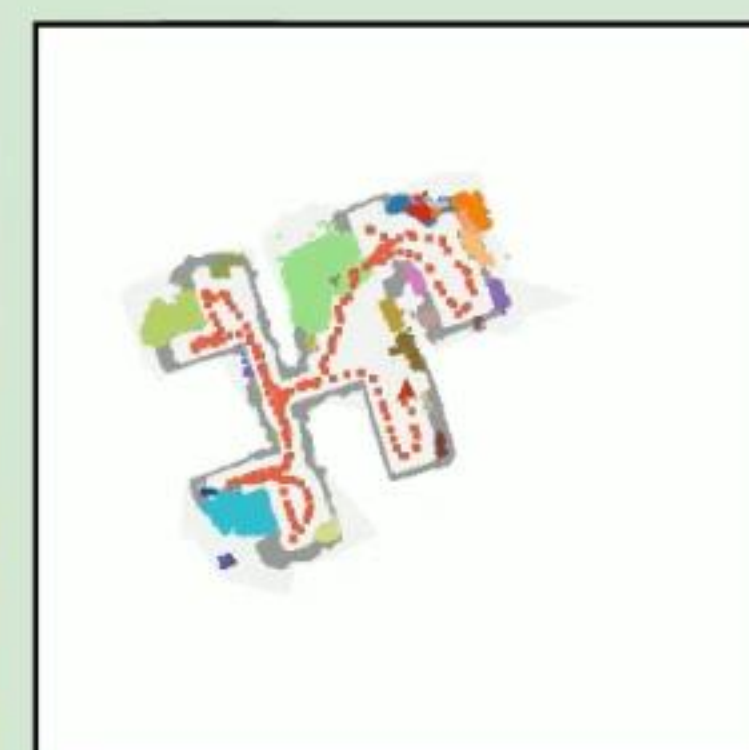
Instance Map

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CoW



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1. GOAT Problem

GOAT System Architecture

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Applications

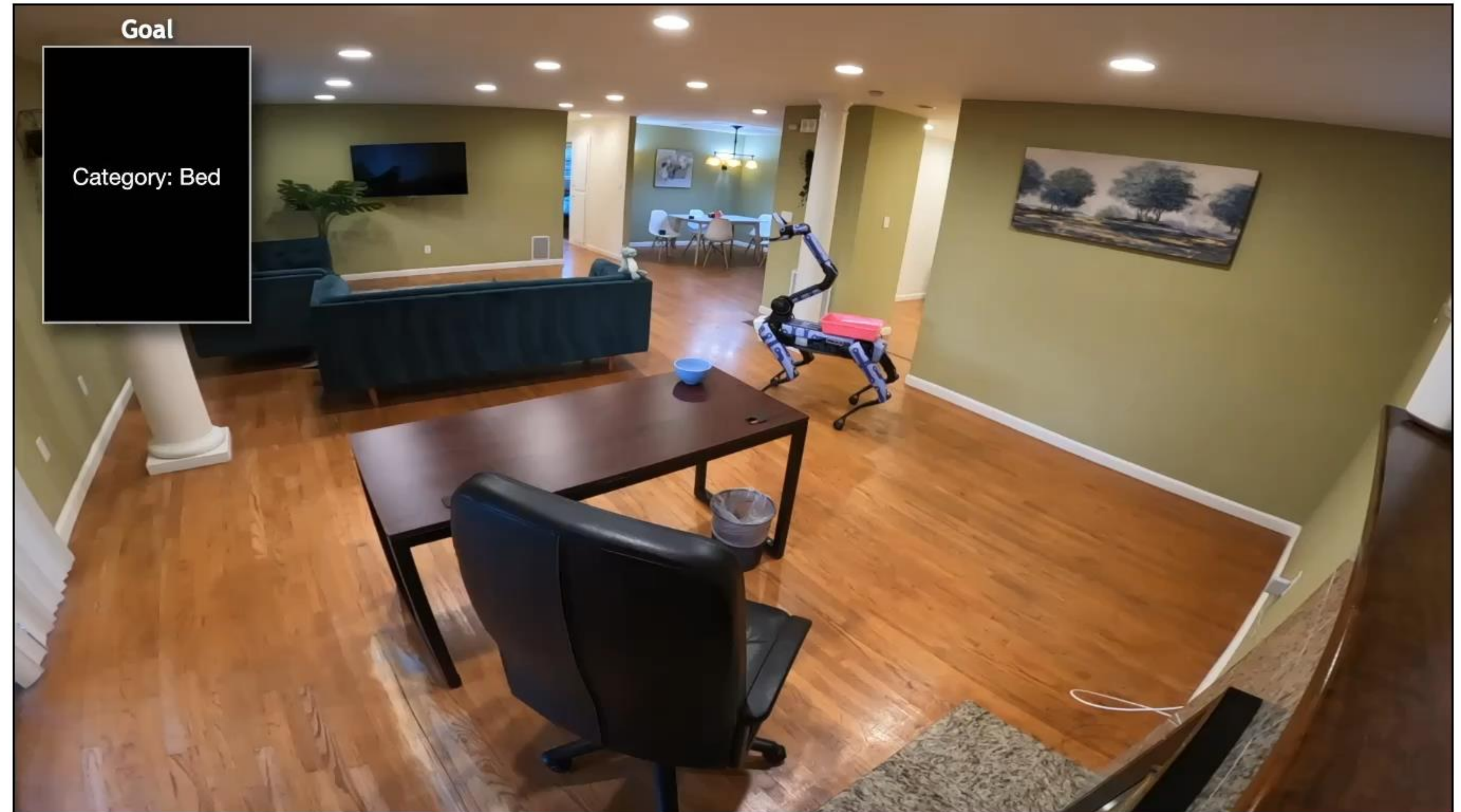
Pick & Place

Social Navigation

Platform Agnostic

Pick & Place

Third-person view



Social Navigation

Observation



Third-person view



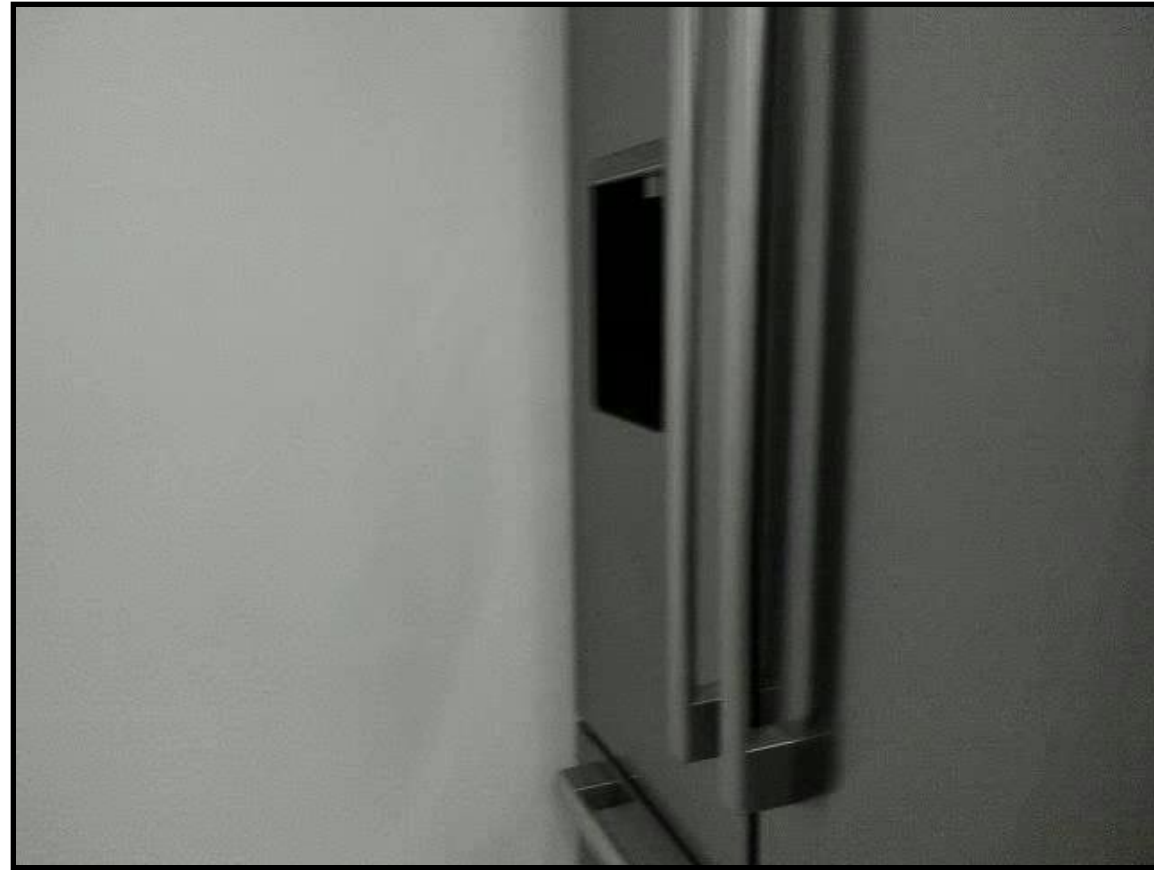
Semantic Map



Robot plans around the dynamic obstacle (person) to go to the refrigerator

Social Navigation

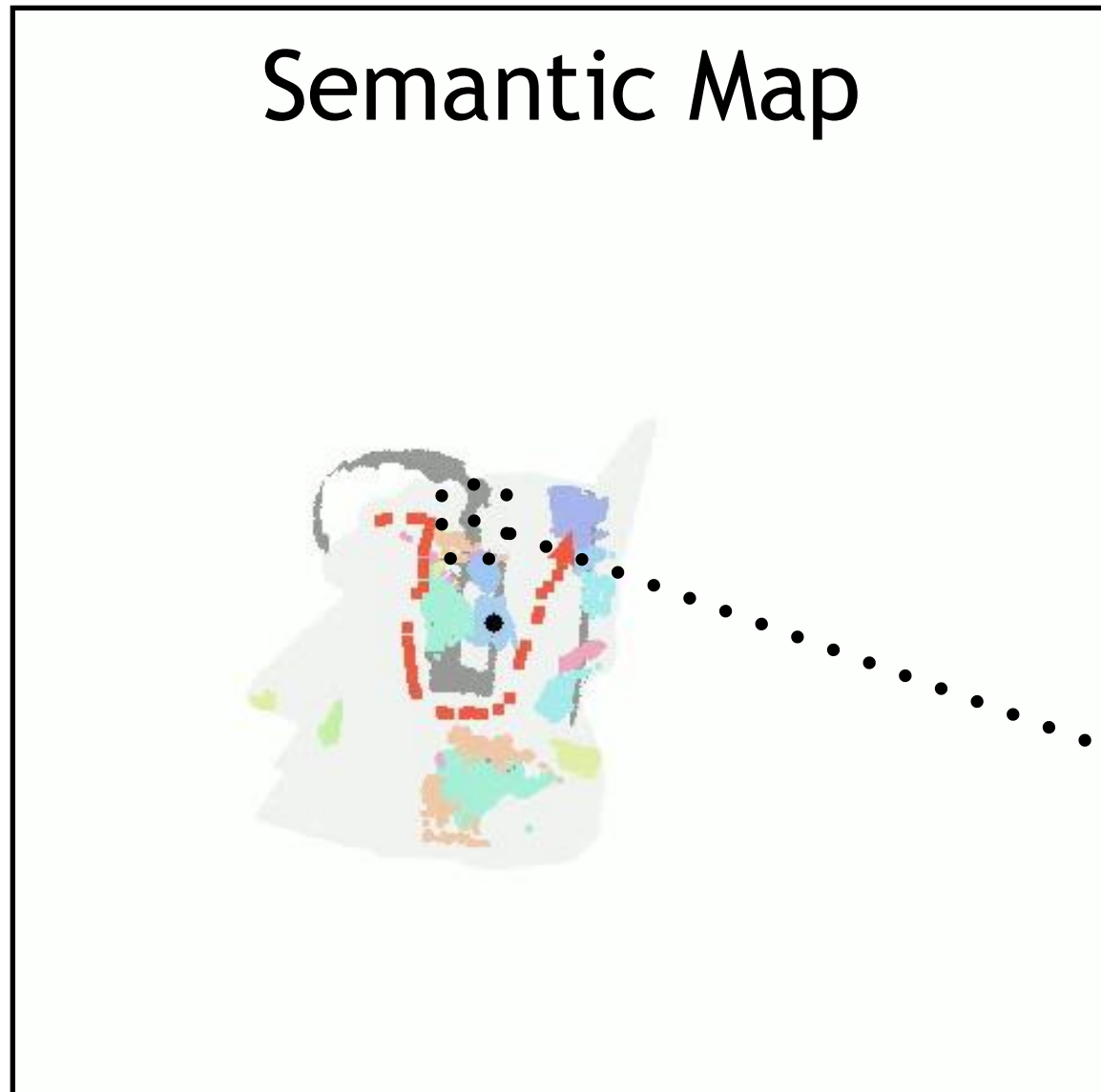
Observation



Third-person view



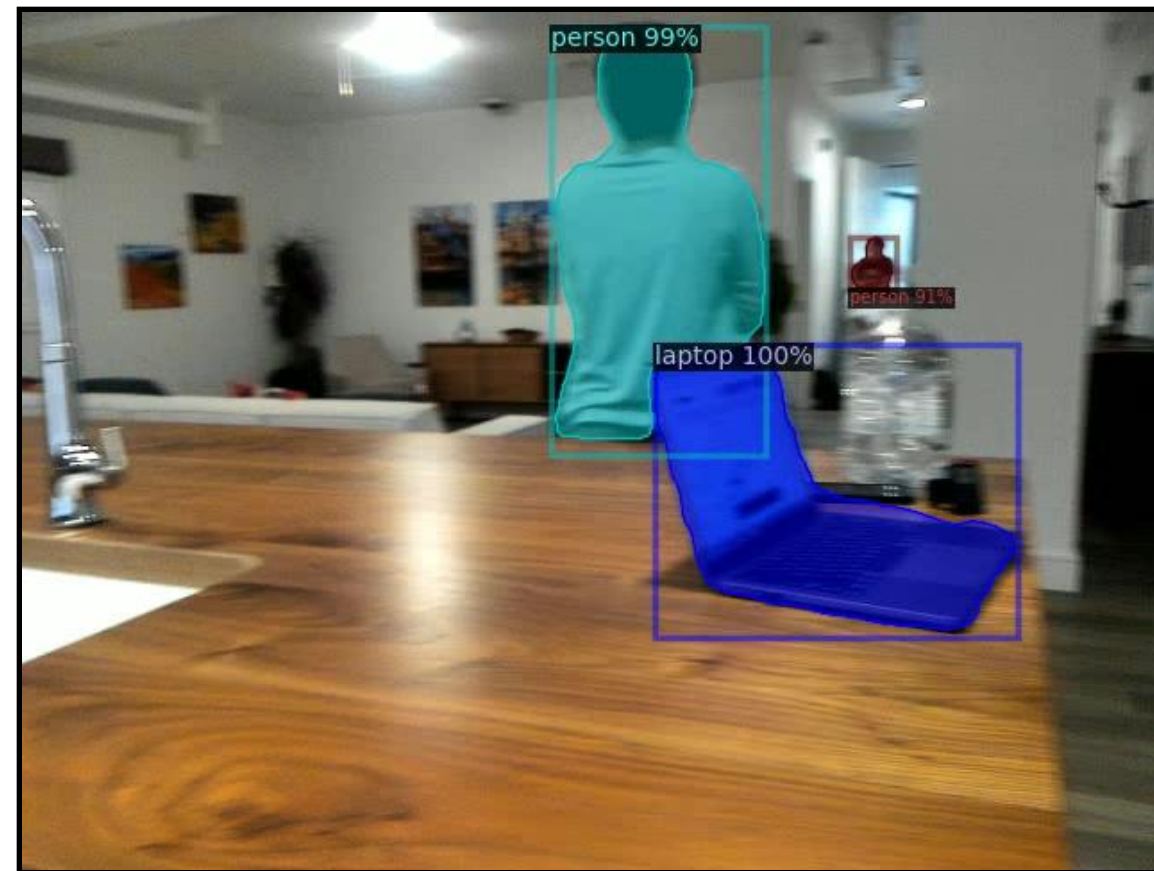
Semantic Map



Robot removes previous location of person from the map

Social Navigation

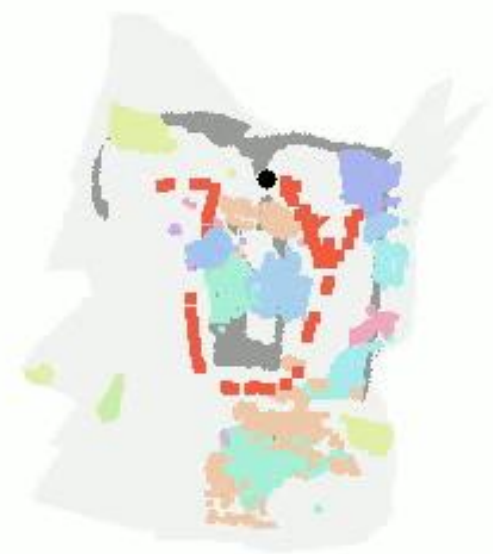
Observation



Third-person view



Semantic Map



Robot follows the person while updating their location

Platform Agnostic

Third-person view

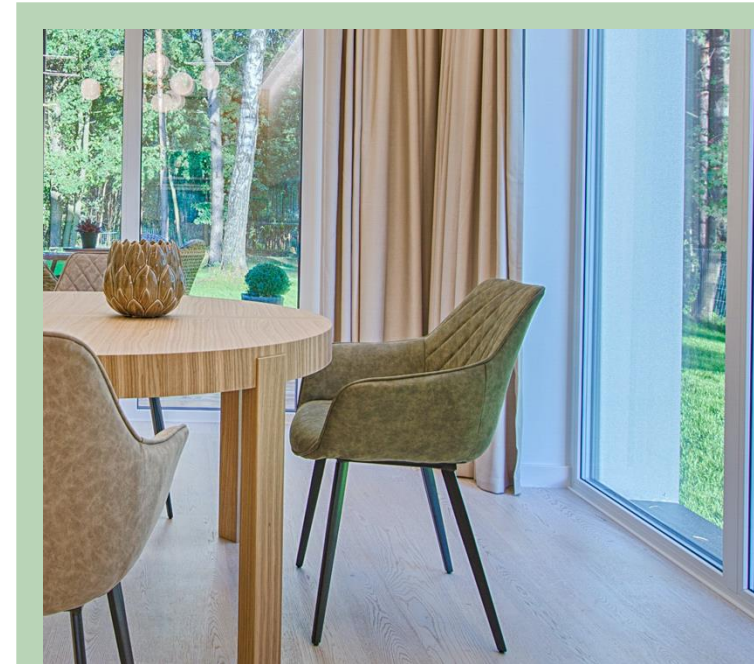


Summary

Universal navigation

- Multimodal

Image



Language

*Find **the fruit basket** on the kitchen counter*

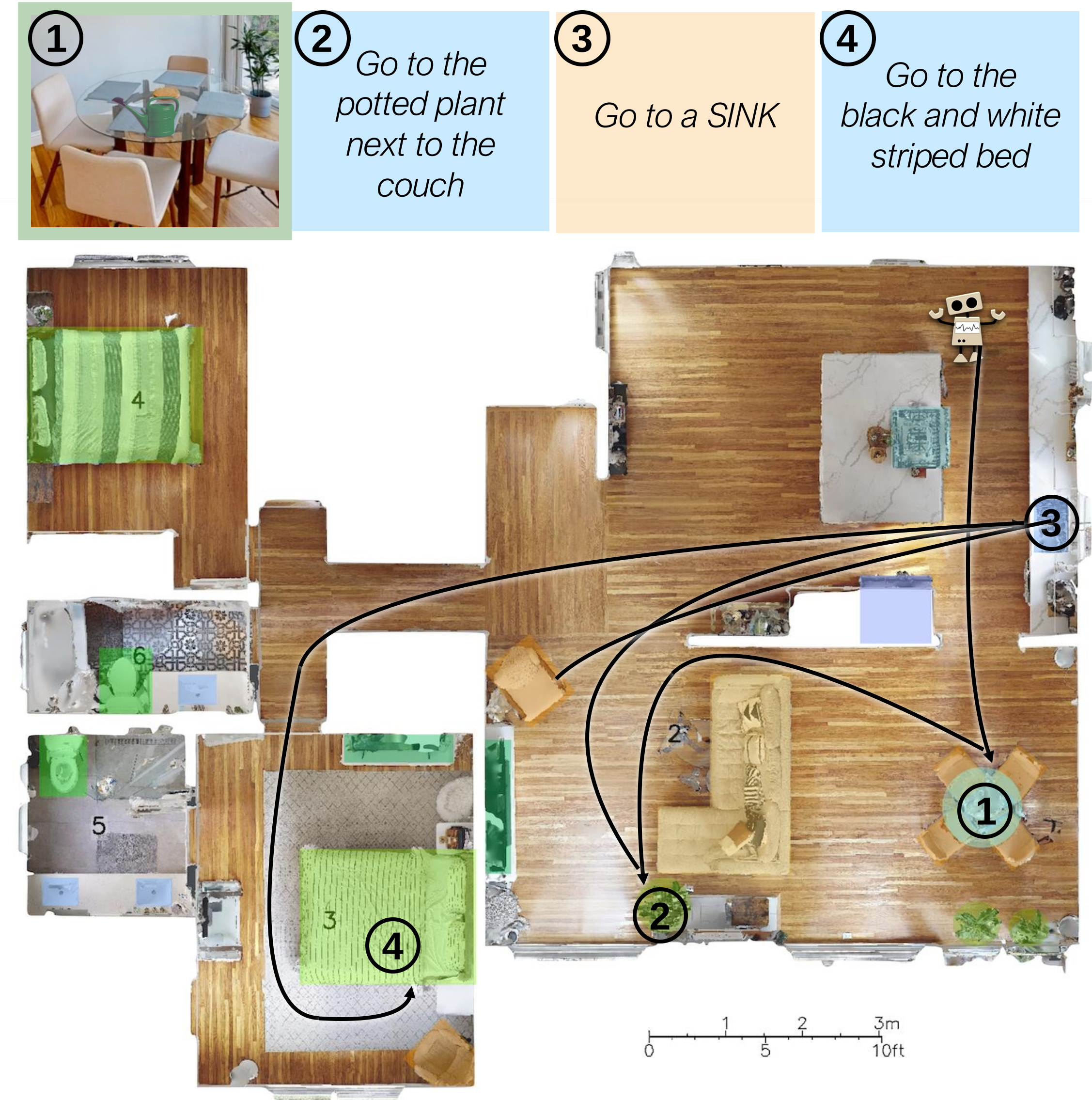
Category

*Bring me a **CUP***

Summary

Universal navigation

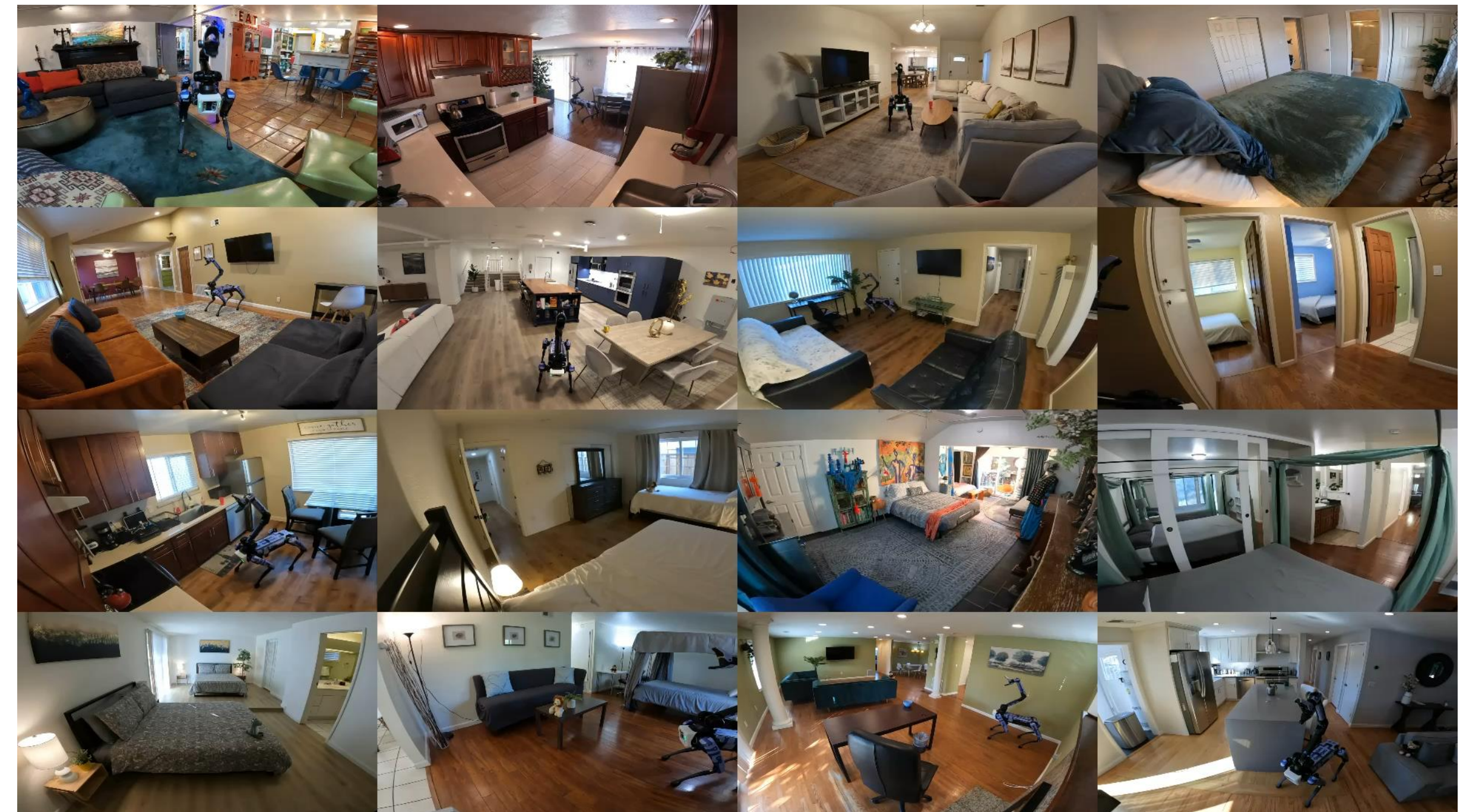
- Multimodal
- Lifelong



Summary

Universal navigation

- Multimodal
- Lifelong
- Unseen environments



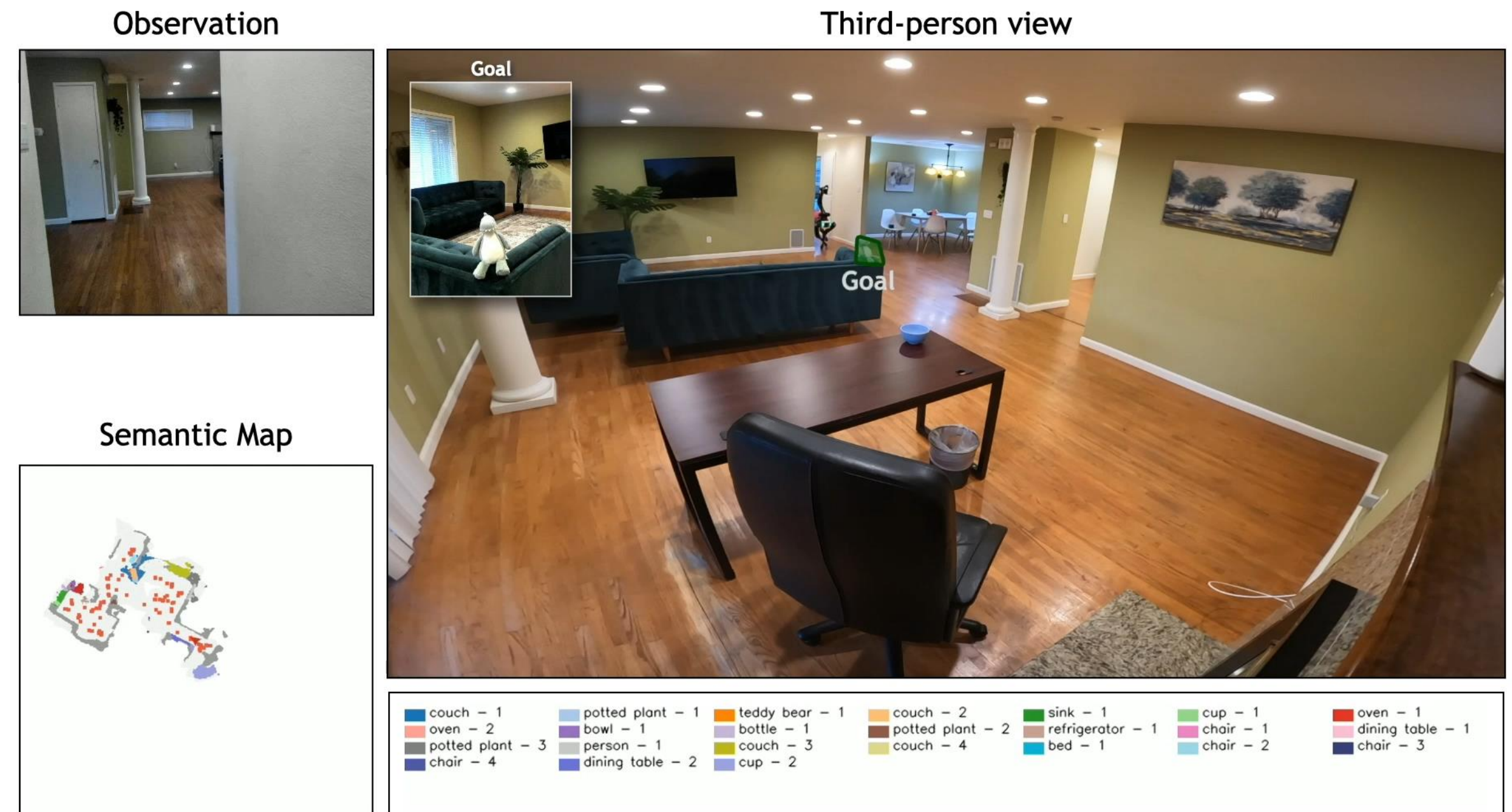
Summary

Universal navigation

- Multimodal
- Lifelong
- Unseen environments

Applications

- Pick & Place



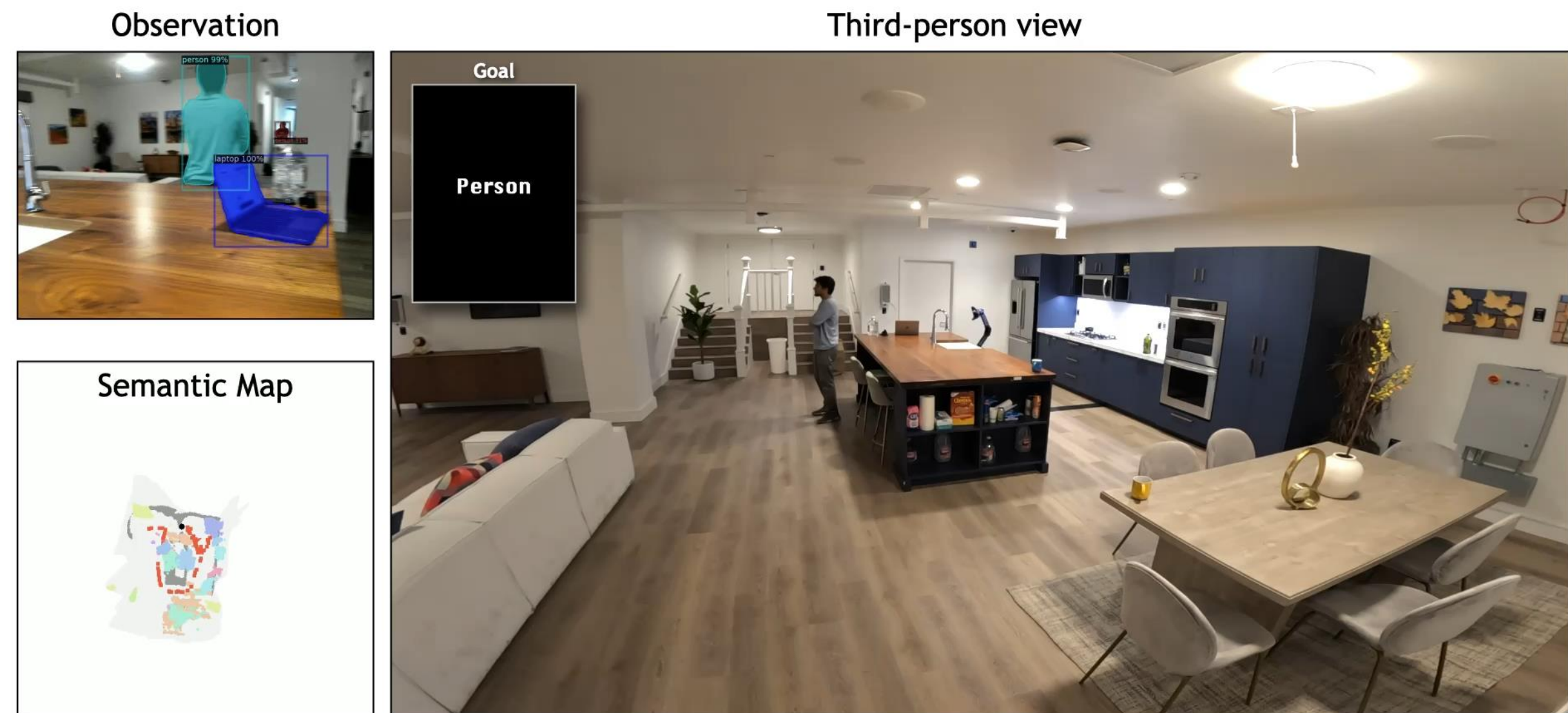
Summary

Universal navigation

- Multimodal
- Lifelong
- Unseen environments

Applications

- Pick & Place
- Social Navigation



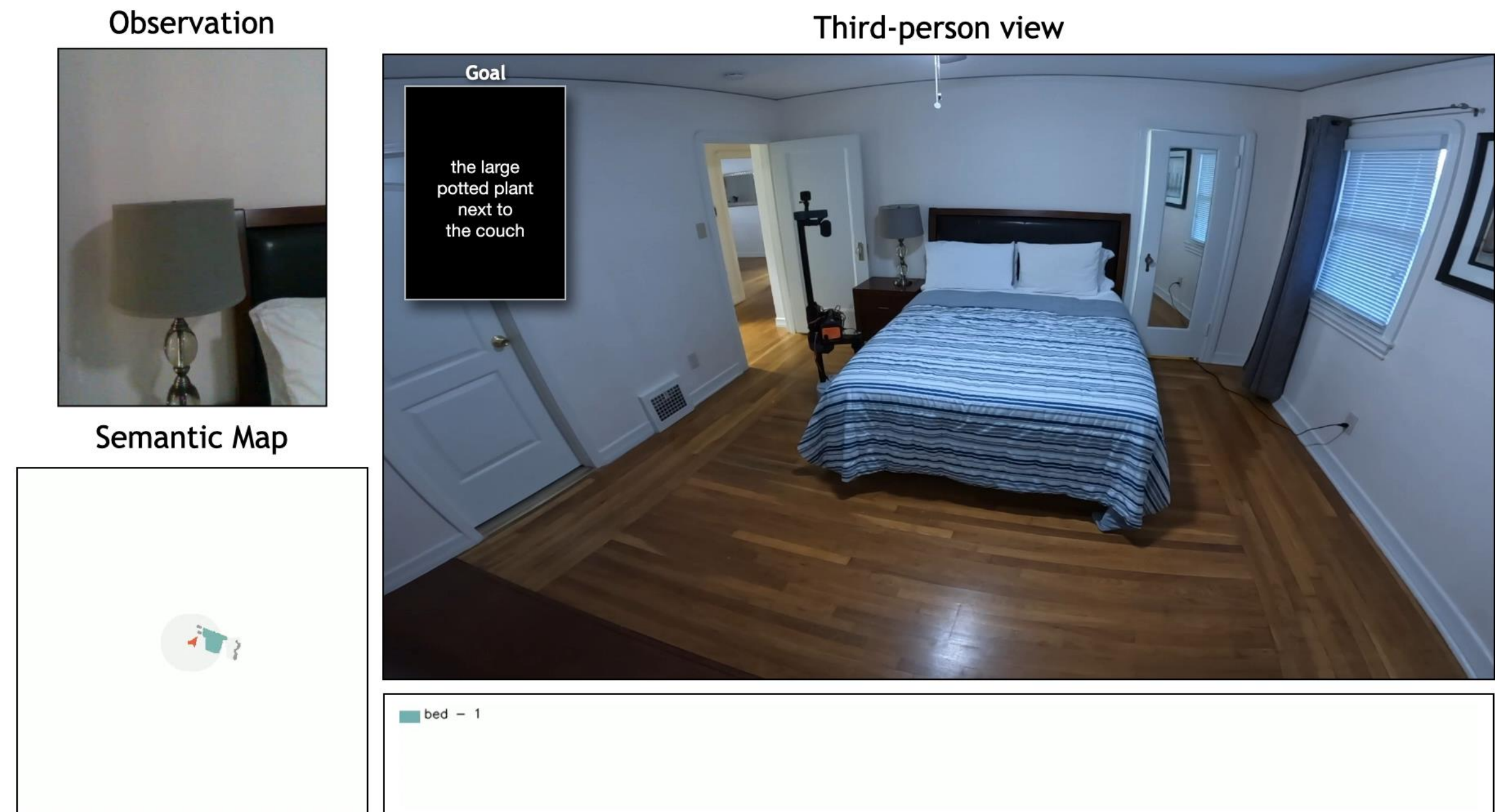
Summary

Universal navigation

- Multimodal
- Lifelong
- Unseen environments

Applications

- Pick & Place
- Social Navigation
- Platform Agnostic





Thank you!

Webpage: <https://theophilegervet.github.io/projects/goat>



Thank you!

